From a Flame to a Fire: Birth of “Deep” Learning

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Spring 2026

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - Shallow learning: why doesn’t it work?
  - Navigating the loss landscape: how to make learning converge
  - Generalizing to the real world: avoiding overfitting and underfitting
  - Programming tutorial

- Assignments (Canvas)
  - Problem set 1 due tomorrow
  - Lab assignment 2 due in 2 weeks

- Questions?

Today's Topics

- NN neglect: 2nd AI winter & retreat to shallow models with engineered features
- NN hope lingers: the unfashionable quest for *depth*
- NNs take over: AlexNet triggers a revolution where 'deep learning' dominates
- Programming tutorial: AlexNet in PyTorch

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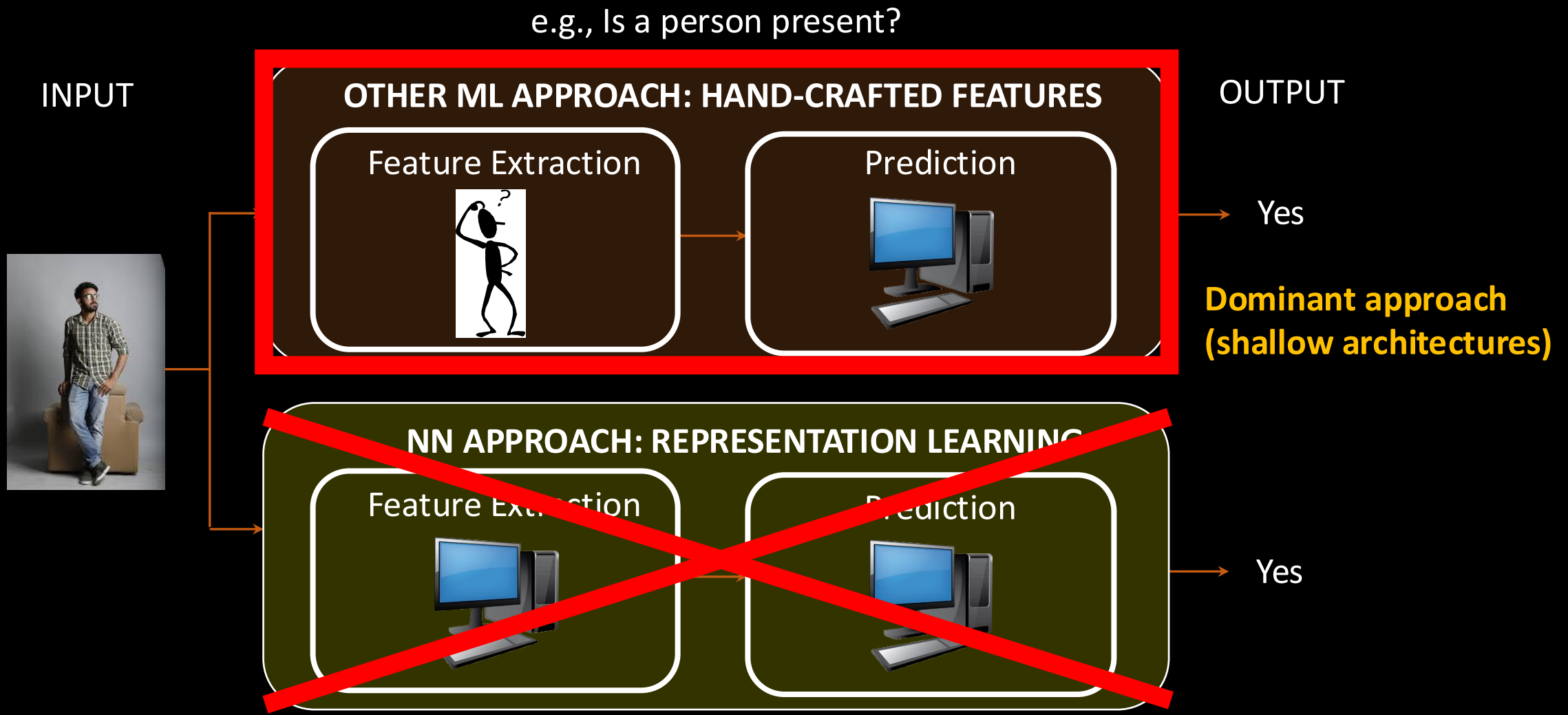
2nd “AI Winter” (late-1980s to late-1990s)



- Dominant, hyped paradigm of rule-based expert systems didn’t deliver
- Funding agencies lost interest
- Researchers left for other topics

https://medium.com/@bergholz/proclaimed-dead-twice-the-ai-winters-5848c7f8a29f

Machine Learning Survived and Sparked New Hope for AI



Machine Learning Survived and Sparked New Hope for AI

Machine Learning, 20, 273–297 (1995)
© 1995 Kluwer Academic Publishers, Boston. Manufactured in The Netherlands.

Support-Vector Networks

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VLADIMIR VAPNIK vlad@neural.att.com
AT&T Bell Labs., Holmdel, NJ 07733, USA

SVMs, in particular, were a key choice

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Shallow vs Deep Architectures

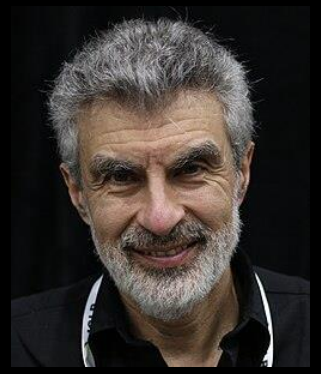
Scaling Learning Algorithms towards AI

Yoshua Bengio (1) and Yann LeCun (2)

(2007; published in the book
Large-scale Kernel Machines)



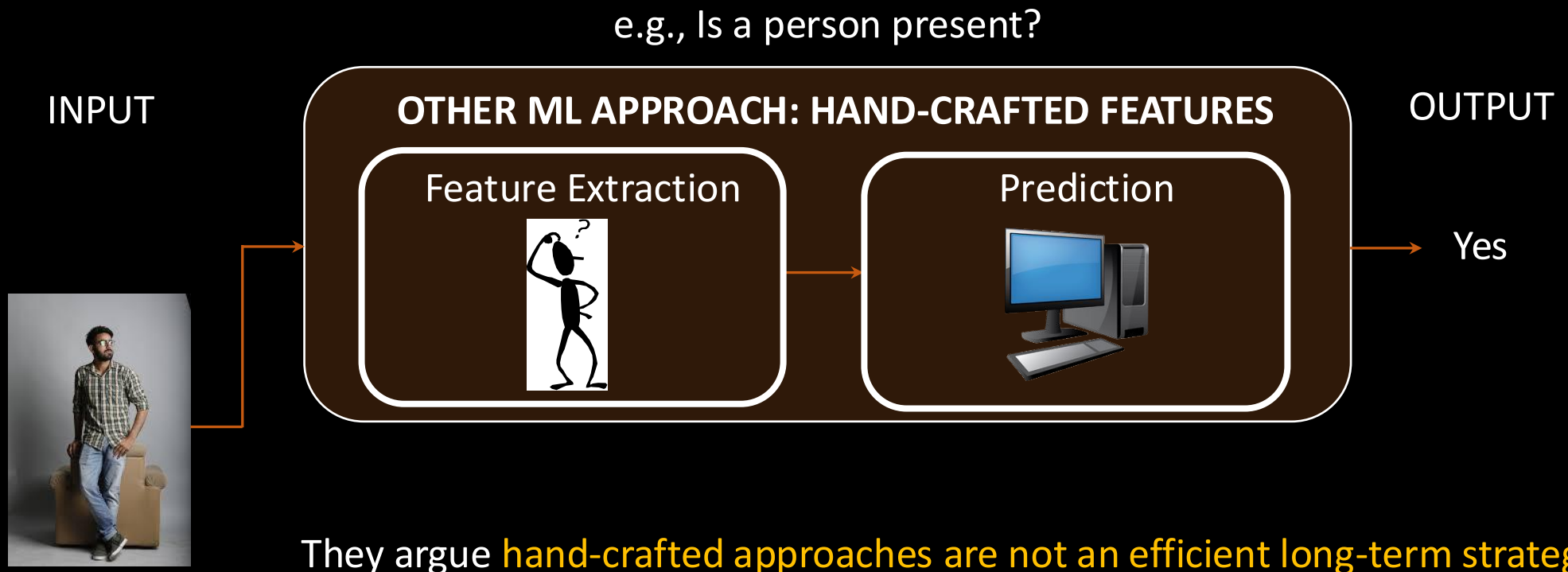
Yann LeCun
1960 – Present
Professor, Former Chief AI Scientist (Meta)



Yoshua Bengio
1964 – Present
Professor, Co-President of LawZero

"We argue that deep architectures… crucial in order to make progress on the kind of complex tasks required for artificial intelligence.

Shallow vs Deep Architectures



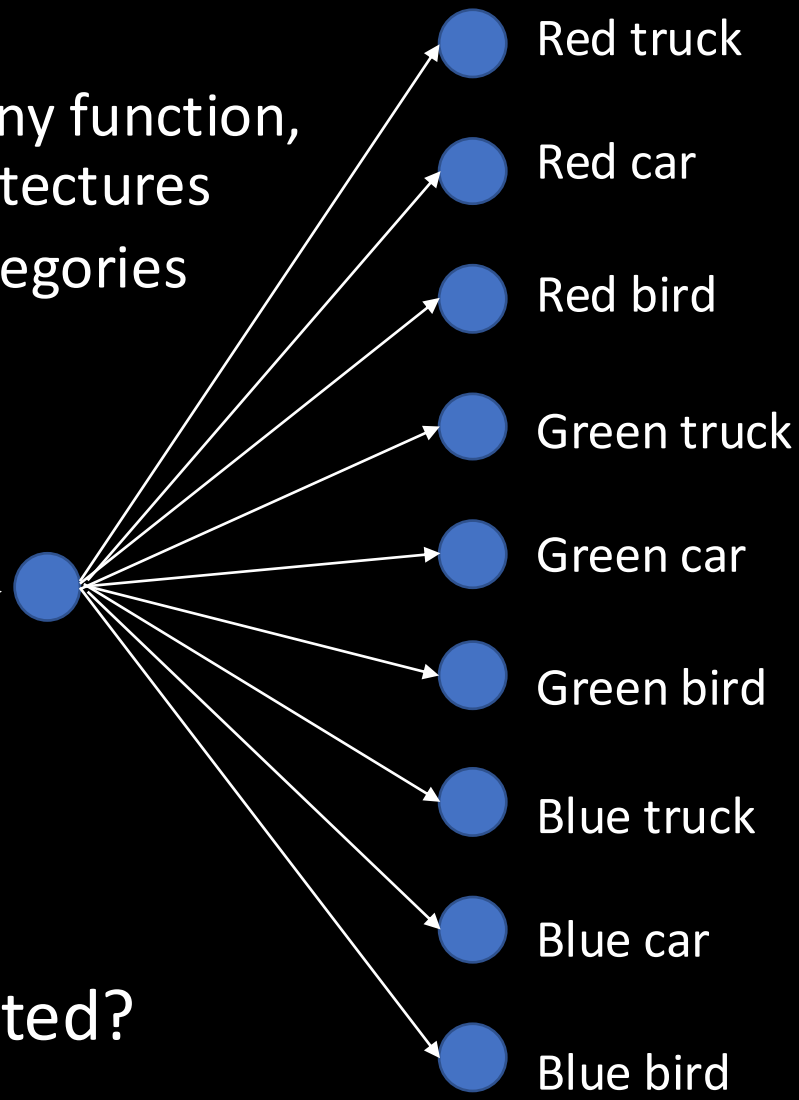
They argue hand-crafted approaches are not an efficient long-term strategy; e.g.,

- "very attractive in the short term because they often yield quick results in making progress on a specific task"
- fundamentally flawed: "If task-specific designs must be devised by hand for each new task, achieving AI will require an overwhelming amount of human effort"

Shallow vs Deep Architectures

- Key insight: while shallow architectures can represent any function, they offer less efficient representations than deep architectures
  - e.g., predict for the image one of the following 9 categories



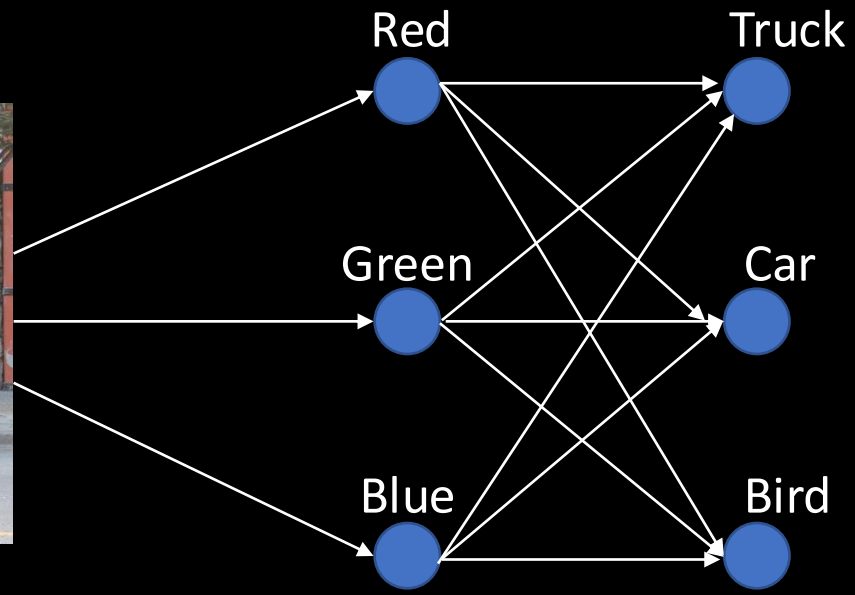


How else could this be represented?

Shallow vs Deep Architectures

- Key insight: while shallow architectures can represent any function, they offer less efficient representations than deep architectures
  - e.g., predict for the image one of the following 9 categories





More efficient representation: capture color and then objects!

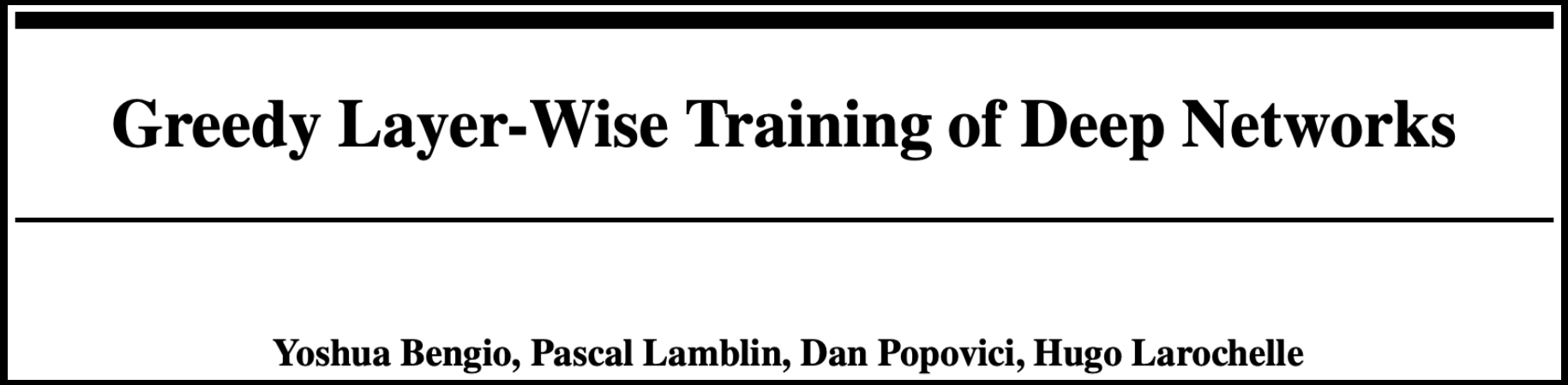
Shallow vs Deep Architectures

- Key insight: while shallow architectures can represent any function, they offer less efficient representations than deep architectures
- Problem: deeper NNs often failed with shallow learning techniques
  - Analogy: what would happen if you scaled an ant to an elephant's size?
  - It would collapse under its own weight (needs different fundamentals, like legs)



Shallow vs Deep Architectures

- Key insight: while shallow architectures can represent any function, they offer less efficient representations than deep architectures
- Problem: deeper NNs often failed with shallow learning techniques
- Renewed hope: after LeCun's LeNet-5 came Bengio's 2006 success



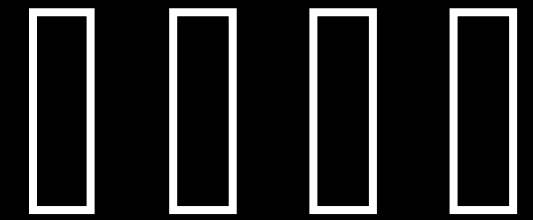
Greedy Layer-Wise Training of Deep Networks

Yoshua Bengio, Pascal Lamblin, Dan Popovici, Hugo Larochelle

(NeurIPS)

Idea: Initialize Parameters Near Good Minima

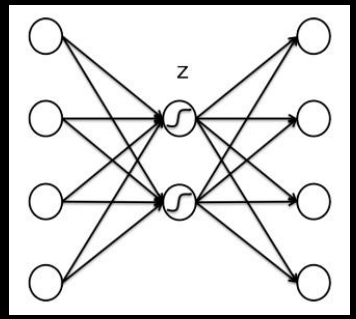
- Approach:
  1. Unsupervised training of each layer sequentially, building depth gradually



Input Train Train Train

  2. Supervised fine-tuning for the target task (add output layer and then train)
- Why unsupervised learning? - lots of data for supervision; e.g.,
  - Autoencoder learning to predict itself, encoding the input followed by decoding into a prediction
  - Given this 620 x 426 input, what would a perfect autoencoder predict:
    - The 264,120 flattened pixels in the input image
  - Intuition behind autoencoders: which is easier to remember
    - **A:** 30, 27, 22, 11, 6, 8, 7, 2
    - **B:** 30, 15, 46, 23, 70, 35, 106, 53, 160, 80, 40, 20, 10, 5





https://lazyprogrammer.me/a-tutorial-on-autoencoders/

    - B; only need a compact set of rules - divide by 2 if even; multiply by 3 and add 1 otherwise

Experiments: “Deep Learning” Holds Promise!

Autoencoder:

	train.	valid.	test
DBN, unsupervised pre-training	0%	1.2%	1.2%
Deep net, auto-associator pre-training	0%	1.4%	1.4%
Deep net, supervised pre-training	0%	1.7%	2.0%
Deep net, no pre-training	.004%	2.1%	2.4%
Shallow net, no pre-training	.004%	1.8%	1.9%

For MNIST, outperforms alternatives of (1) a shallow network, (2) training *without* greedy pre-training, and (3) pre-training greedily with *supervised* learning

Vision in 2007 paper for deep learning (stay tuned for 2020s):

"... one should not consider the large variety of [AI] tasks separately, but as different aspects of a more general problem: that of learning the basic structure of the world, as seen say through the eyes and ears of a growing animal or a young child. This is an instance of multi-task learning… after training such a system on a large variety of tasks in the AI-set, the system may generalize to a new task from only a few labeled examples. We hypothesize that many tasks in the AI-set may be built around common representations..."

Despite Progress, Resistance to NNs Remained

Yann LeCun's famous letter to CVPR organizers about a 2012 paper submission:
(*Paper ratings: "Definitely Reject," "Borderline", "Weakly Reject")*

"… I was very sure that this paper was going to get good reviews because:

1) it has two simple and generally applicable ideas for segmentation ("purity tree" and "optimal cover");
2) it uses no hand-crafted features (it's all learned all the way through. Incredibly, this was seen as a negative point by the reviewers!)
3) it beats all published results on 3 standard datasets for scene parsing;
4) it's an order of magnitude faster than the competing methods.

If that is not enough to get good reviews, I just don't know what is."

"*Scene Parsing with Multiscale Feature Learning, Purity Trees, and Optimal Covers*" rejected by CVPR but accepted by ICML'12

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AlexNet Model (Named After 1rst Author)

ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky
University of Toronto
kriz@cs.utoronto.ca

Ilya Sutskever
University of Toronto
ilya@cs.utoronto.ca

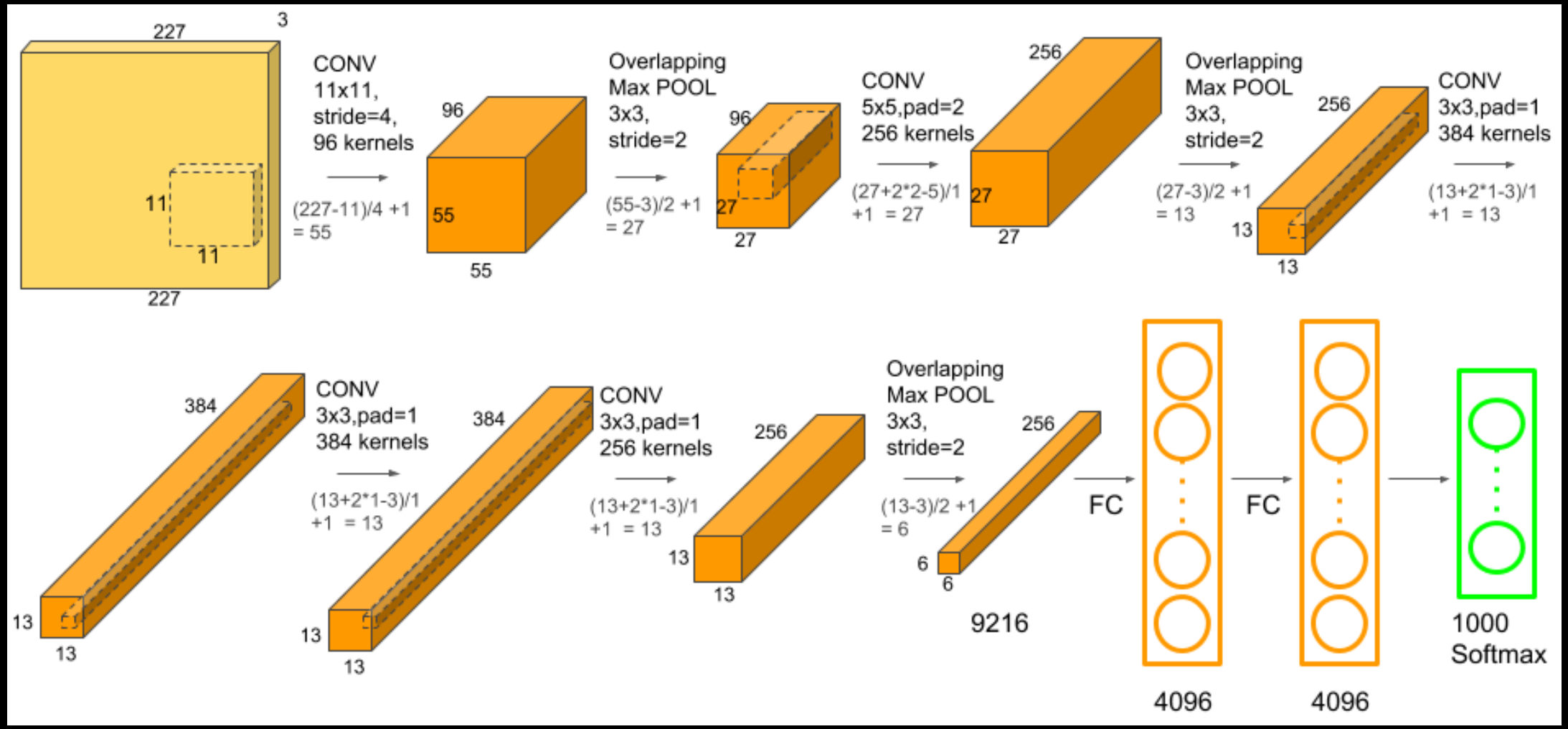
Geoffrey E. Hinton
University of Toronto
hinton@cs.utoronto.ca

(2012, NeurIPS)



1947 – Present
Professor; Google Brain
2024 Nobel Prize winner

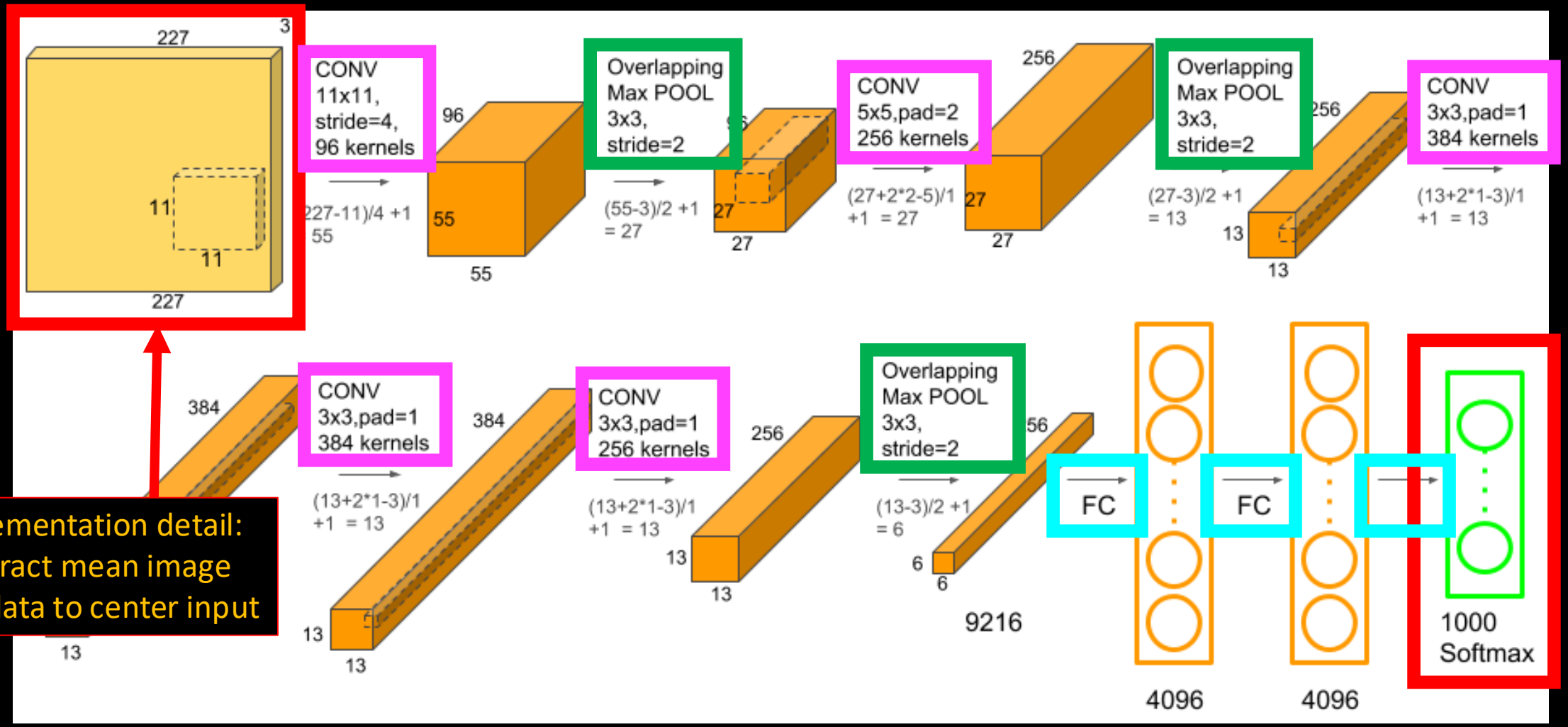
AlexNet Architecture: Going “Deeper”



https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png

AlexNet Architecture

Input: RGB image resized to fixed input size
Output: 1000 class probabilities (sums to 1)
Convolutional layers: 5 layers
Pooling Layers: 3 layers
Fully-connected layers: 3 layers



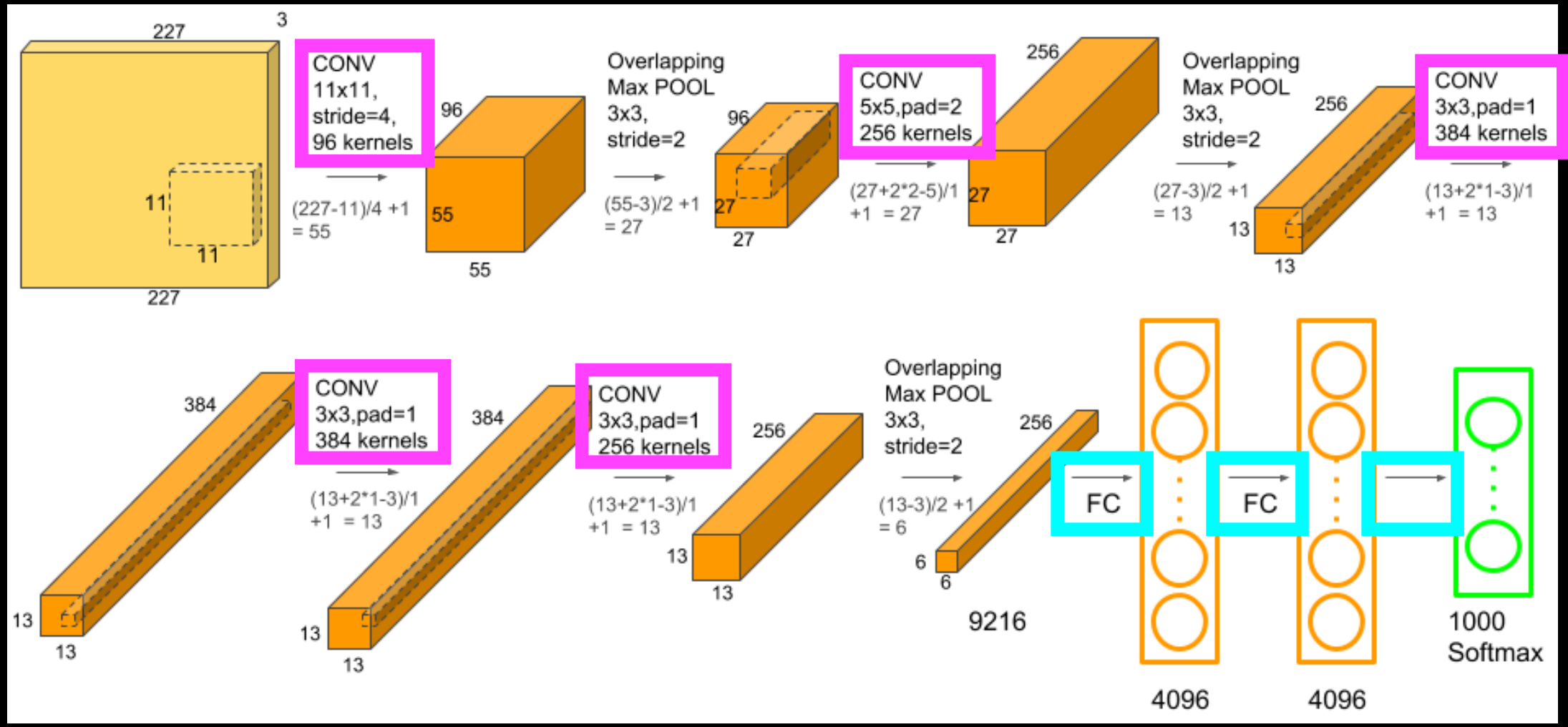
Implementation detail: subtract mean image from data to center input

https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png

AlexNet Architecture

Altogether, 60 million model parameters must be learned!

How many layers have model parameters that need to be learned?



https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png

Premise: Large Capacity Model Necessary for Real-World Complexity

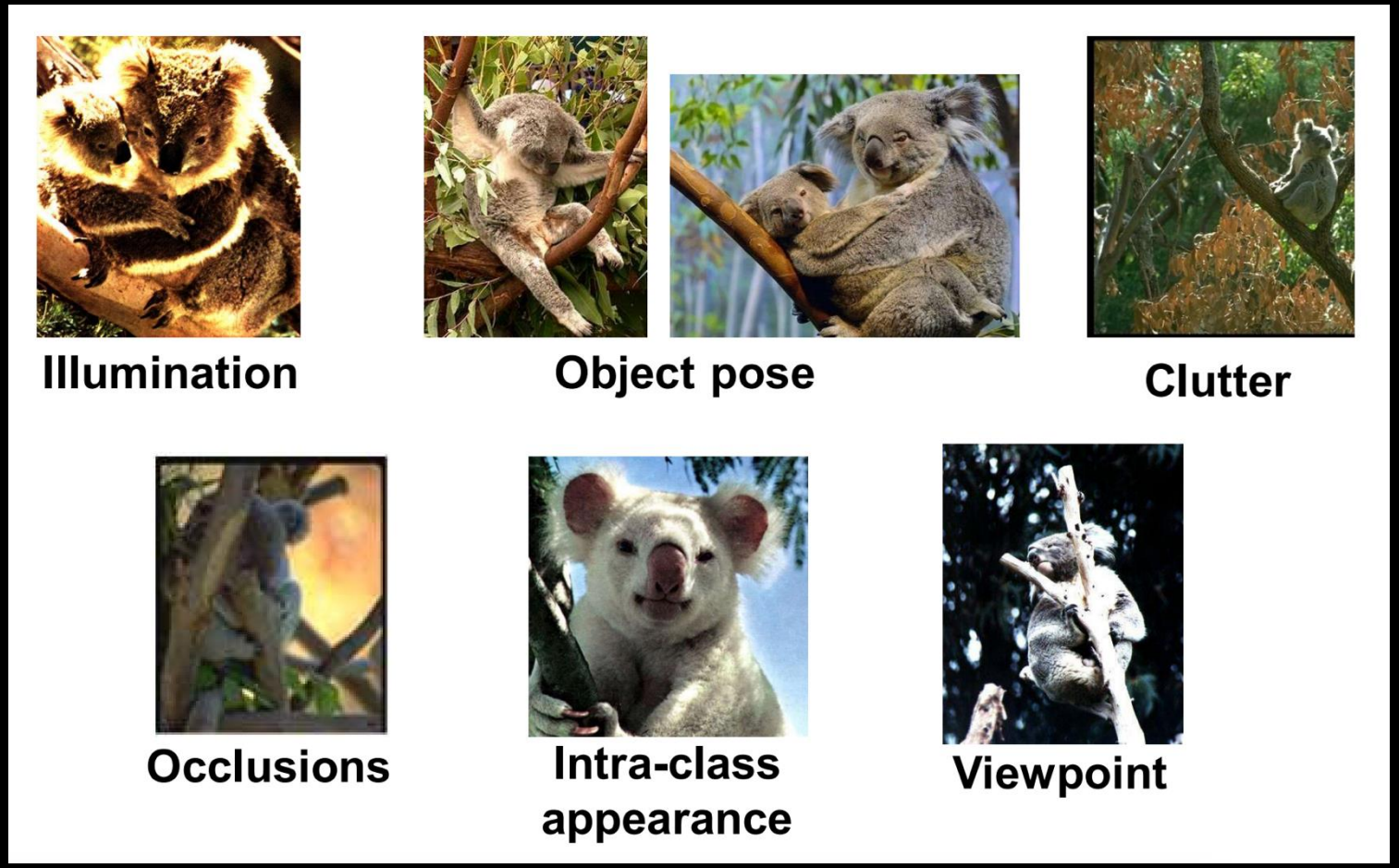
- Task: predict category from 1000 options
- Dataset: ~1.5 million images of objects in natural backgrounds
- Source: images scraped from search engines, such as Flickr, and labeled by crowdworkers
- Evaluation metric: % correct (top-1 and top-5 predictions)



Deng et al. ImageNet: A Large-Scale Hierarchical Image Database. 2009

Premise: Large Capacity Model Necessary for Real-World Complexity

Consider complexity for just one object category:



Source: Kristen Grauman

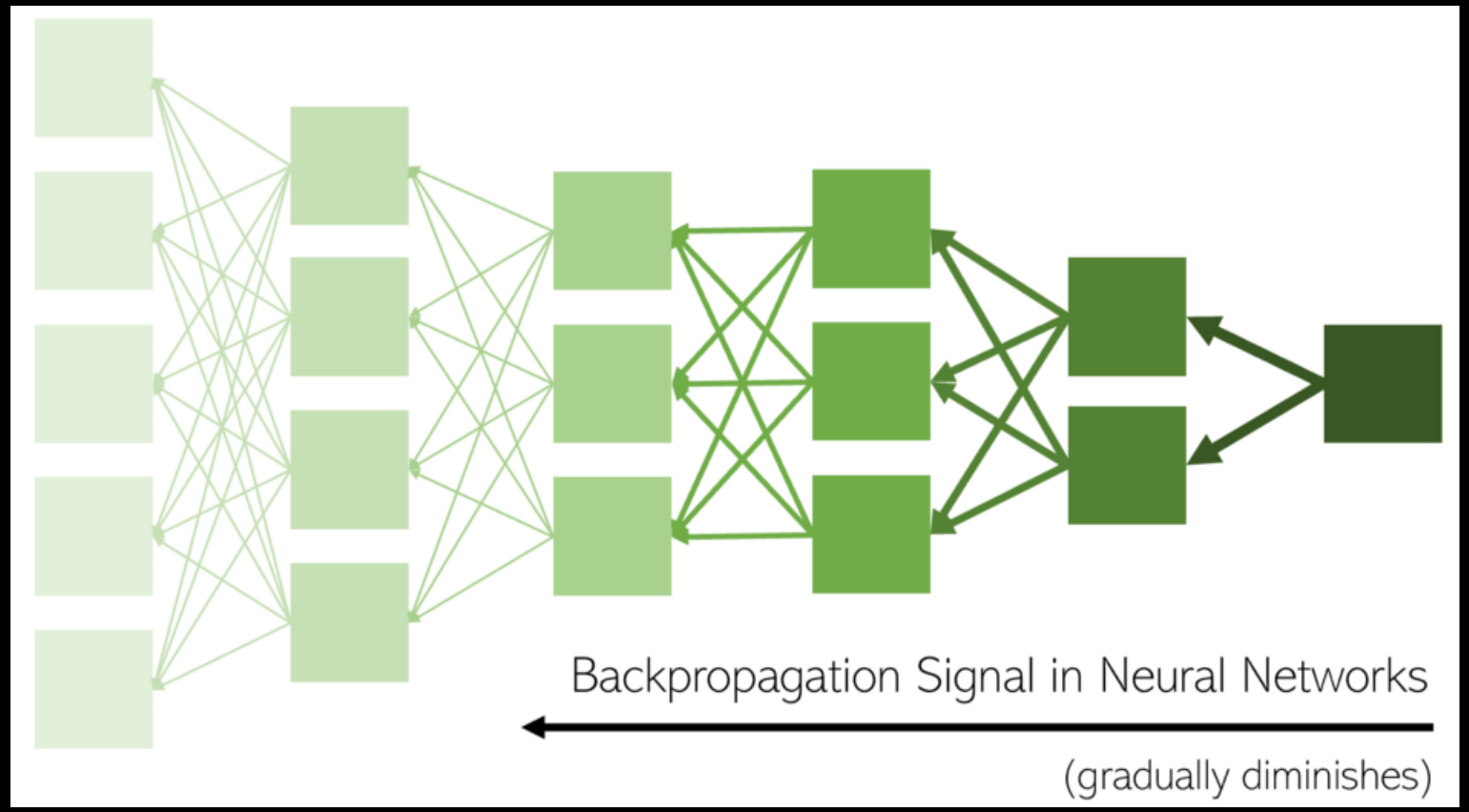
AlexNet Training: What Could Go Wrong?

- Learning is not possible: vanishing gradients
- Learning doesn’t generalize: overfitting
- Learning is not practical: training takes too long to converge

AlexNet Training: Key Ideas For Deep Learning

- Enable learning: use non-saturating activation functions
- Prevent overfitting: incorporate regularization methods
- Make training feasible: speed it up with better hardware

Recall Vanishing Gradient Problem

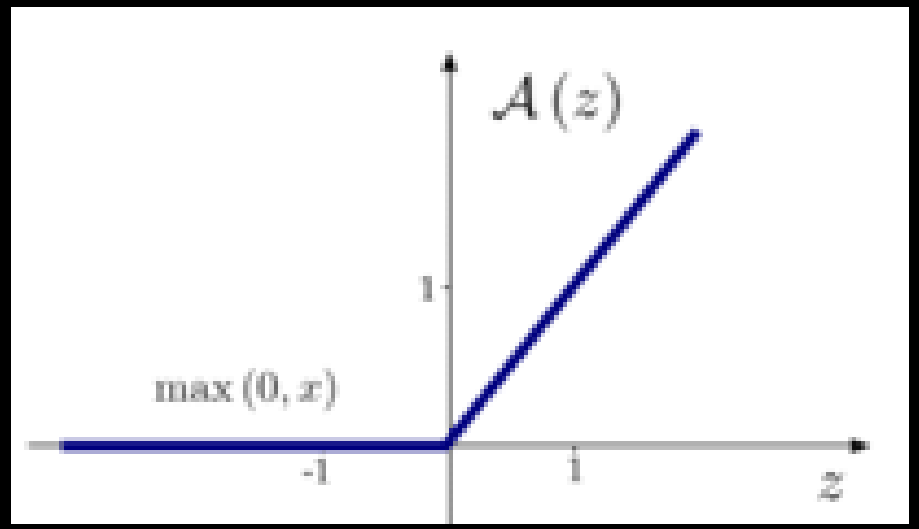


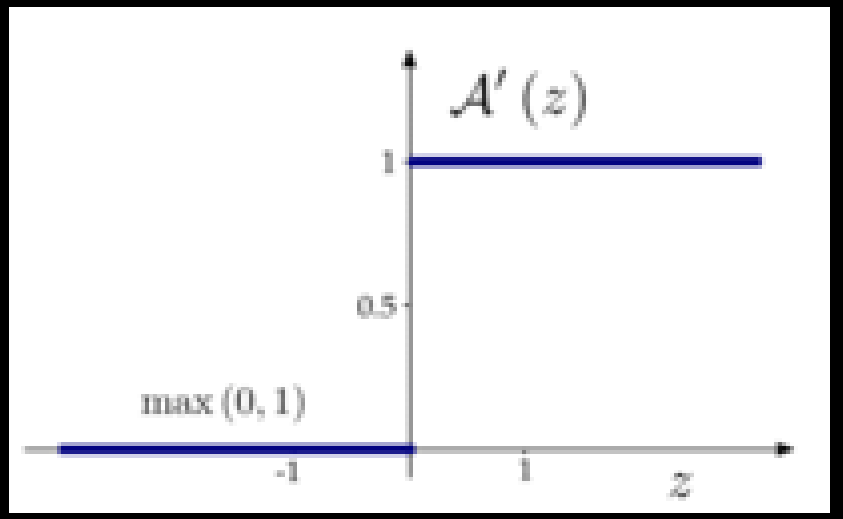
Garbage In = Garbage Out : Earlier layers struggle to get gradients to learn something useful, yet are the foundation for later layers to make predictions

https://towardsdatascience.com/batch-normalization-the-greatest-breakthrough-in-deep-learning-77e64909d81d

Idea: Employ an Activation Function that Preserves Gradients During Backpropagation

ReLU Activation Function





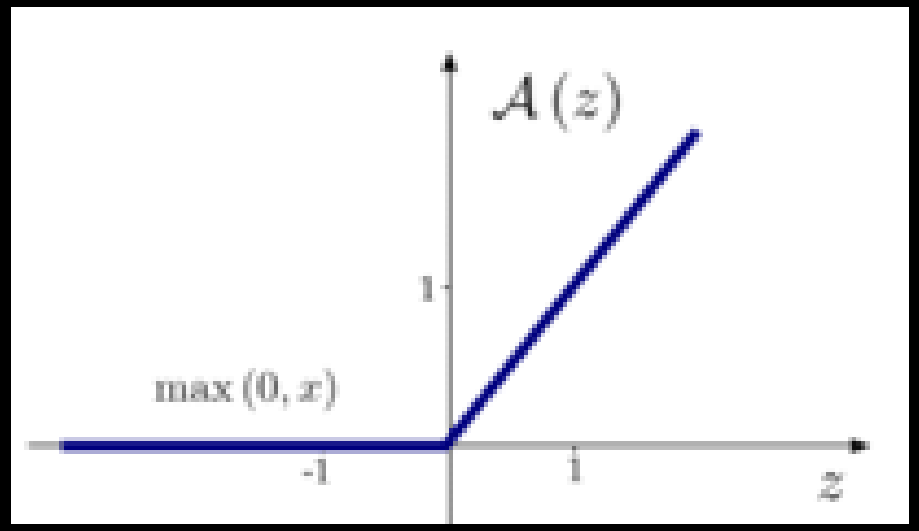
What are possible gradient values?

Note: ReLU was previously used in NNs including by Fukushima for his 1980 Neocognitron and a 1969 precursor as well as Hinton in 2010 with RBMs

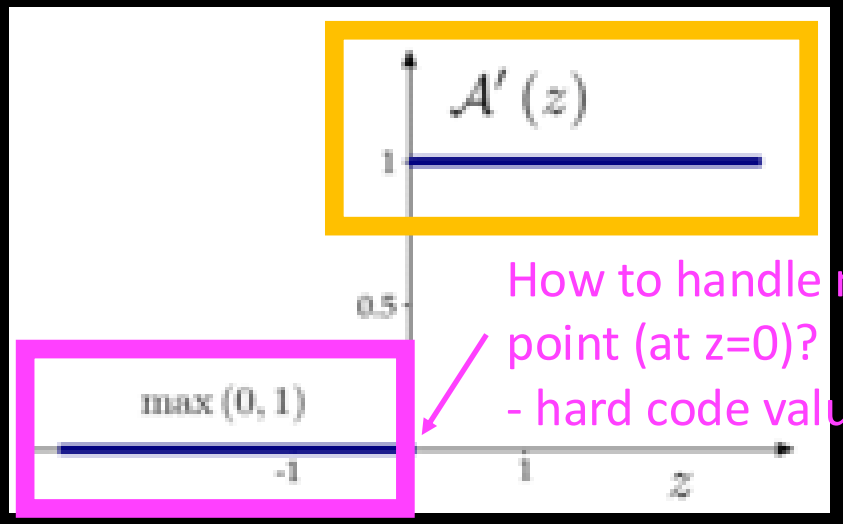
Masi et al. Journal of the Mechanics and Physics of Solids. 2021

Idea: Employ an Activation Function that Preserves Gradients During Backpropagation

ReLU Activation Function



Issues arise when ***many*** neurons lack a gradient; i.e., "dead" (analogy: brain damage)



How to handle non-differentiable point (at z=0)?
- hard code value (i.e., 0 or 1)

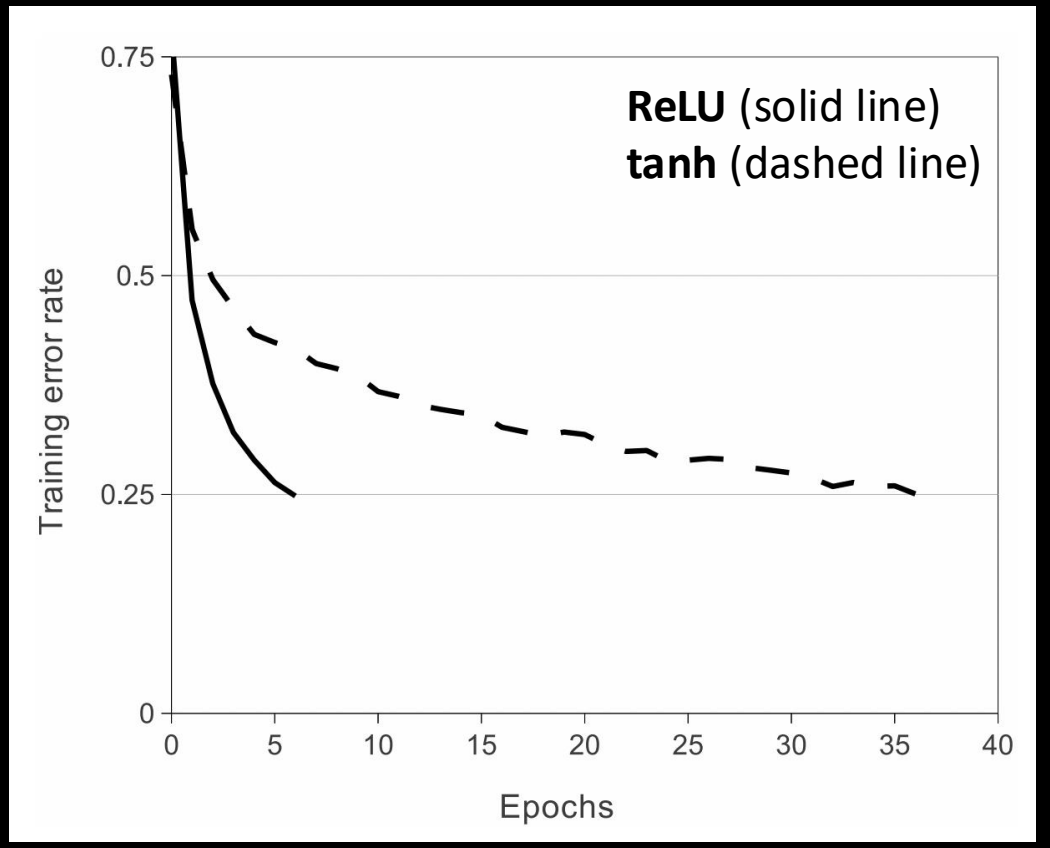
- Gradient won't vanish: i.e., 1x1x1x…
- Fast to compute! - 0 or 1

Masi et al. Journal of the Mechanics and Physics of Solids. 2021

Validating Experimental Analysis

- Dataset: CIFAR-10
- Architecture: 4-layer convolutional network
- Evaluation metric: % correct

What is the key finding?



ReLU learns much faster than tanh, with the latter unsuitable for learning!

(e.g., ReLU is 6x faster in achieving 25% error rate)

Krizhevsky et al. ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 2012.

AlexNet Training: Key Ideas For Deep Learning

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- Prevent overfitting: incorporate regularization methods
- Make training feasible: speed it up with better hardware

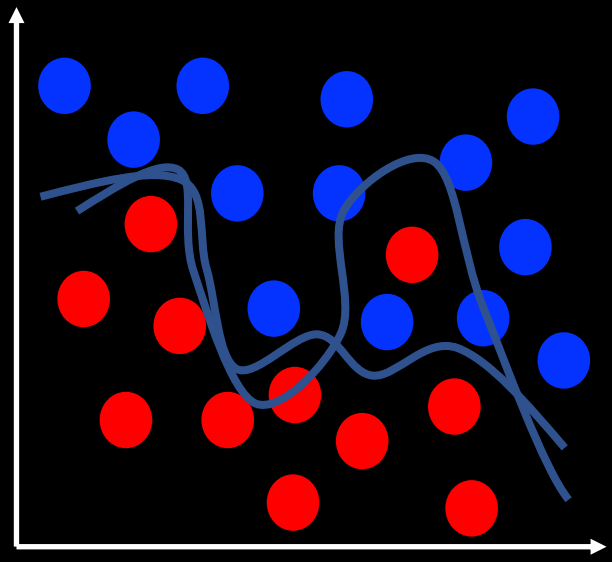
Regularization Methods

- Recall: regularization is “any modification we make to a learning algorithm that is intended to reduce its generalization error.”- Goodfellow book

- Two approaches leveraged by AlexNet: data augmentation & dropout

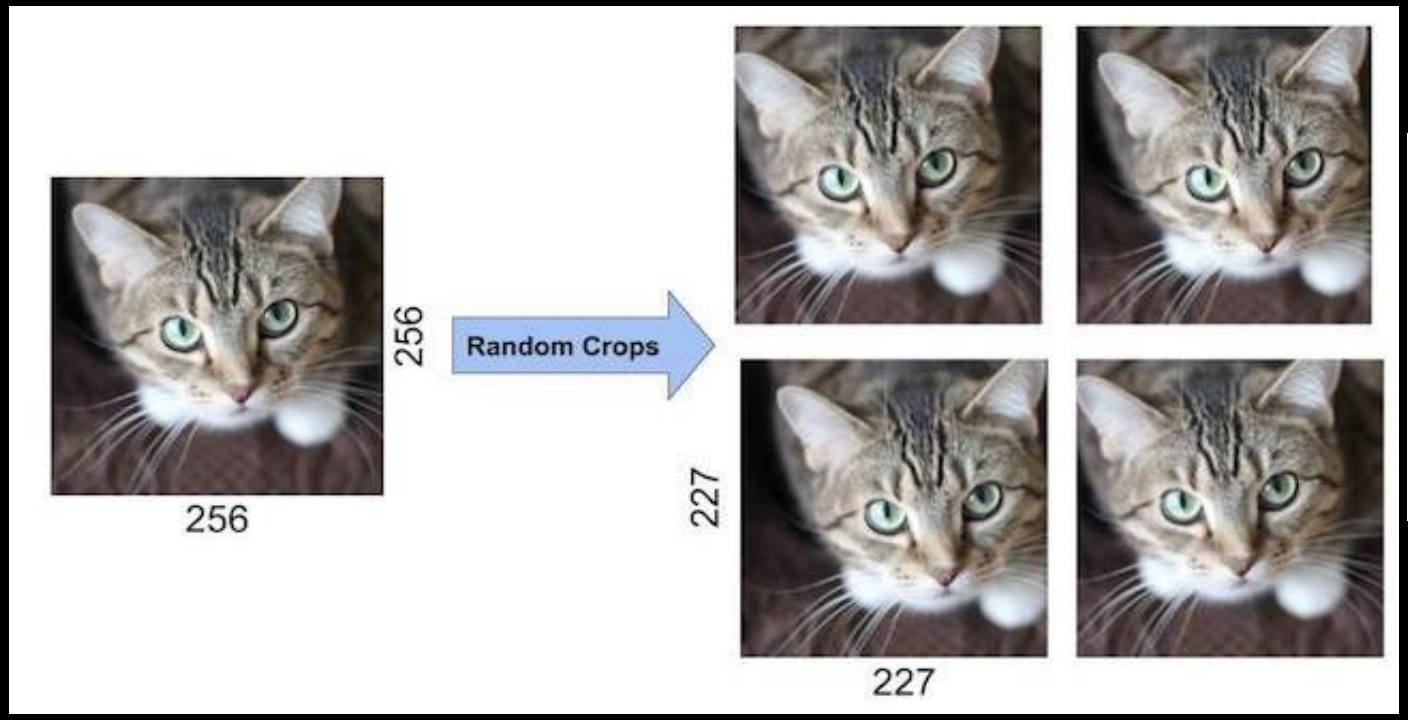
Data Augmentation: Intuition

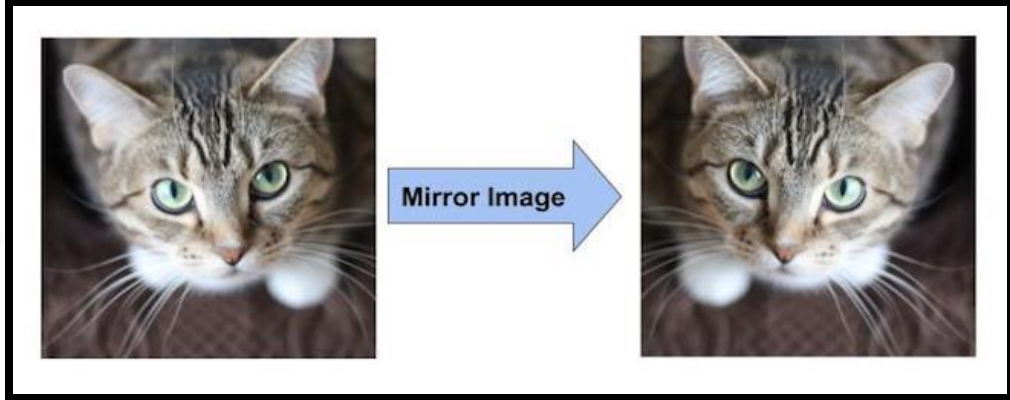
Adding training data



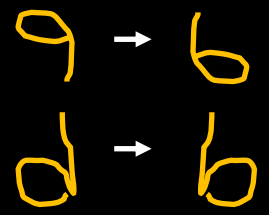
Data Augmentation: Approach

- Random patches and their mirror images (2048x more data)





Generally, need to ensure augmentation scheme aligns with target application:



- Adjust RGB channels (using PCA-based method)

https://learnopencv.com/understanding-alexnet/

Dropout

(2012, arXiv)

Improving neural networks by preventing co-adaptation of feature detectors

G. E. Hinton*, N. Srivastava, A. Krizhevsky, I. Sutskever and R. R. Salakhutdinov
Department of Computer Science, University of Toronto,
6 King's College Rd, Toronto, Ontario M5S 3G4, Canada

Dropout: Idea



Use ensemble



Dropout: Idea

- Using ensemble reduces probability for making a wrong prediction
- Suppose:
  - n classifiers for binary classification task
  - Each classifier has same error rate ε
  - Classifiers are independent (not true in practice!)
  - Probability mass function indicates the probability of error from an ensemble:

Number of classifiers

Classifier error rate

$$P(y \geq k) = \sum_{k}^{n} \binom{n}{k} \varepsilon^{k} (1-\varepsilon)^{n-k} = \varepsilon_{ensemble}$$

Error probability

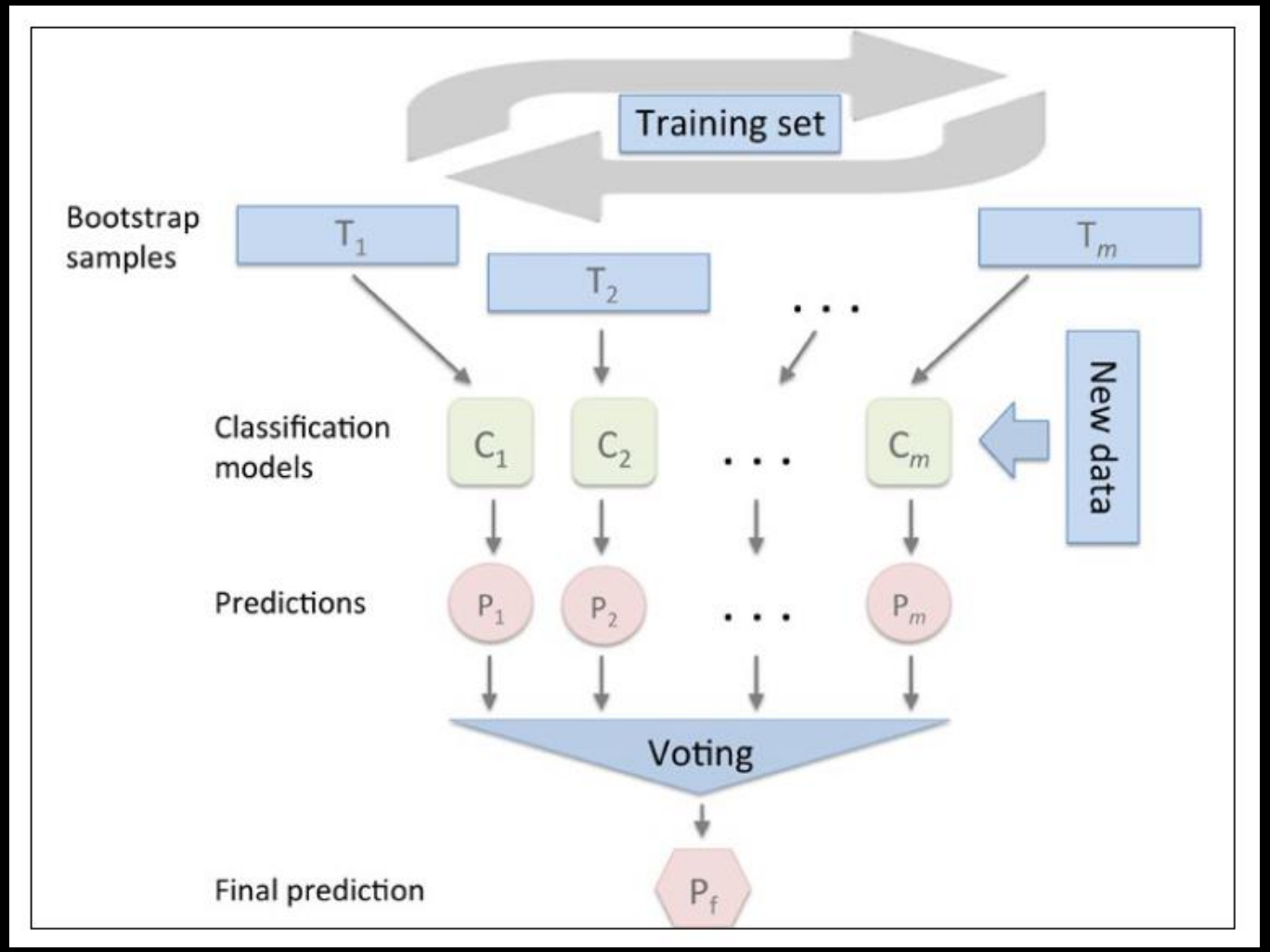
ways to choose *k* subsets from set of size *n*

- e.g., n = 11, ε = 0.25; k = 6: probability of error is ~0.034 which is much lower than probability of error from a single algorithm (0.25)

Dropout: Precursor

Bootstrap Aggregation (1994)

Train algorithm repeatedly on different random subsets of the training set



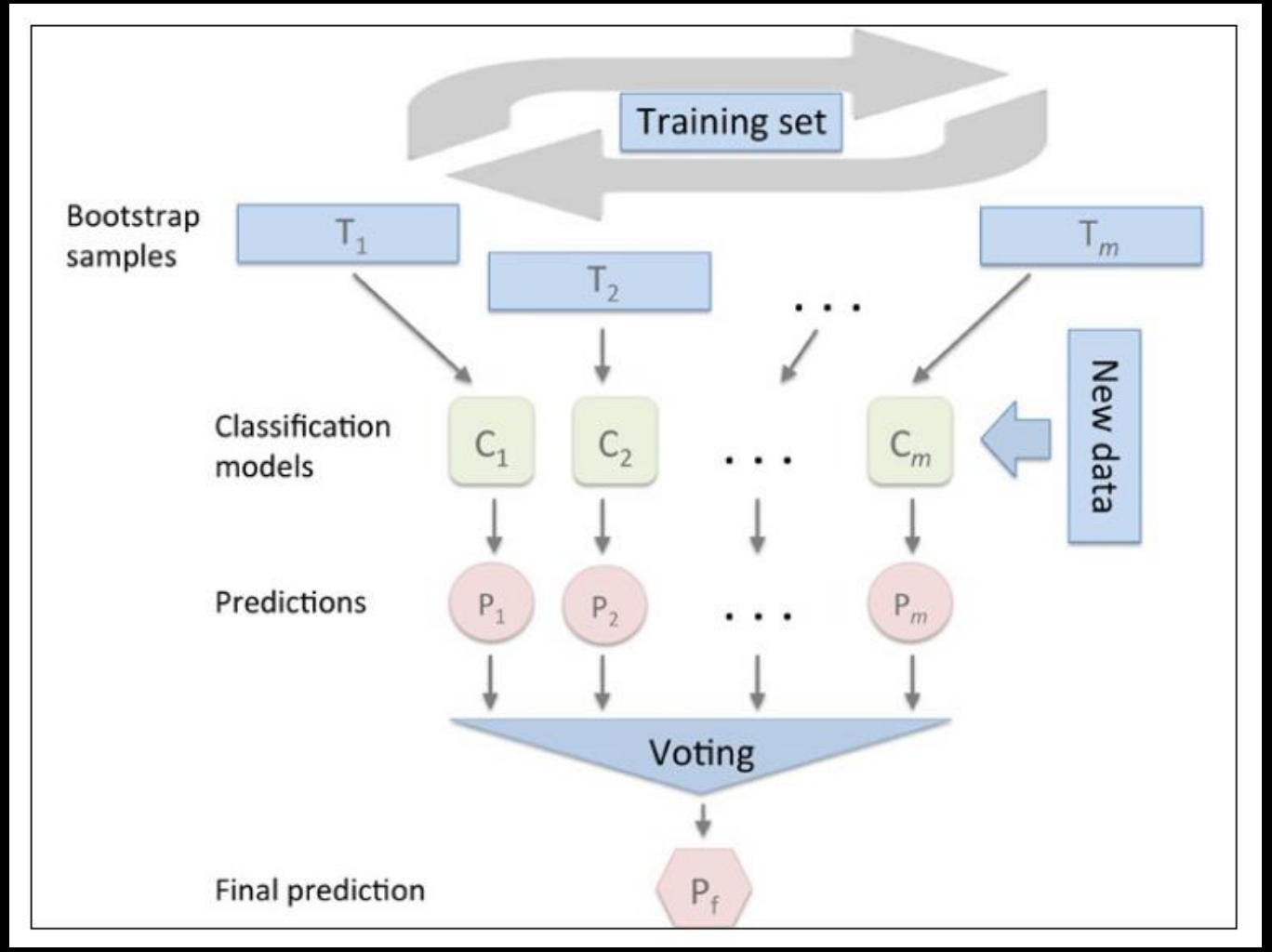
Raschka & Mirjalili, Python Machine Learning.

Dropout: Precursor

Train algorithm repeatedly on different random subsets of the training set

Why is bagging a poor approach for neural networks?

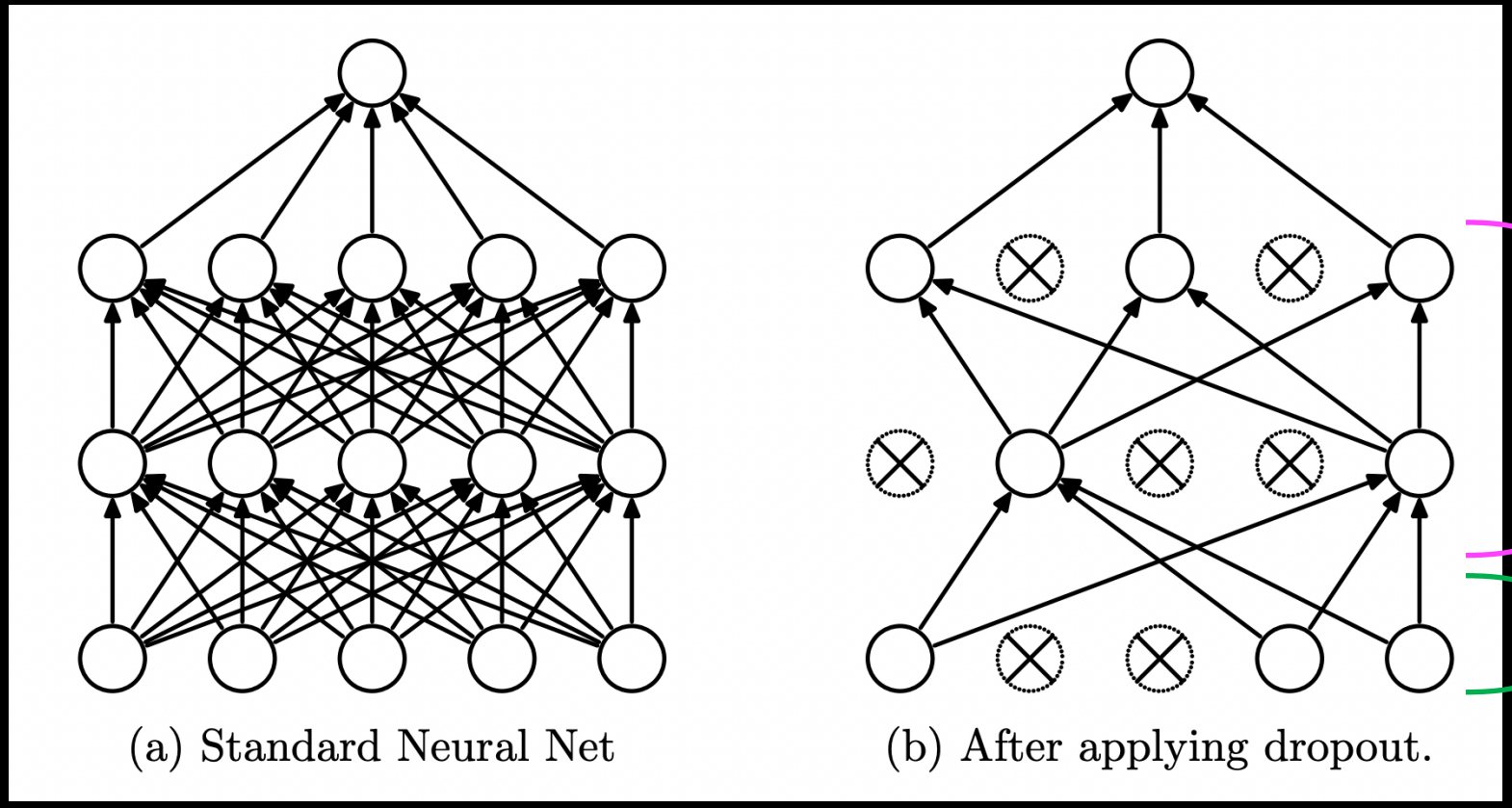
- Finding optimal hyperparameters for each architecture is time-consuming
- Applying multiple neural networks is often infeasible since the models require lots of memory and are computationally expensive to run



Raschka & Mirjalili, Python Machine Learning.

Dropout: Approach

- Approximates bagging with dropout during training so different sub-models in the network are trained with different training data



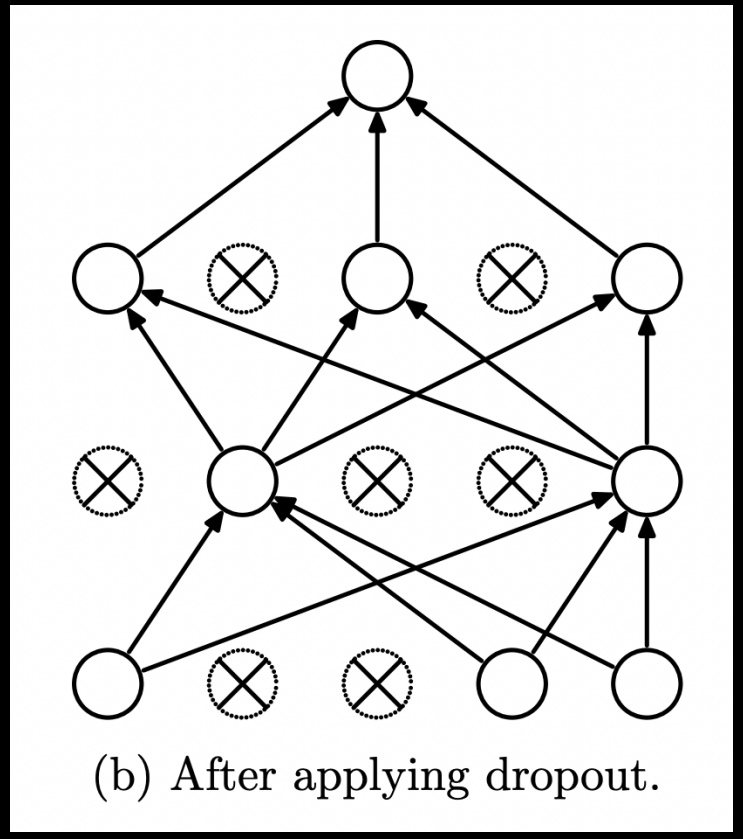
Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

Dropout: Approach

- Approximates bagging with dropout during training so different sub-models in the network are trained with different training data

For training, the forward pass and backpropagation run only through the sub-network (with a different dropout per minibatch).

What might happen to loss curves?
- Bouncier since the underlying network continuously changes



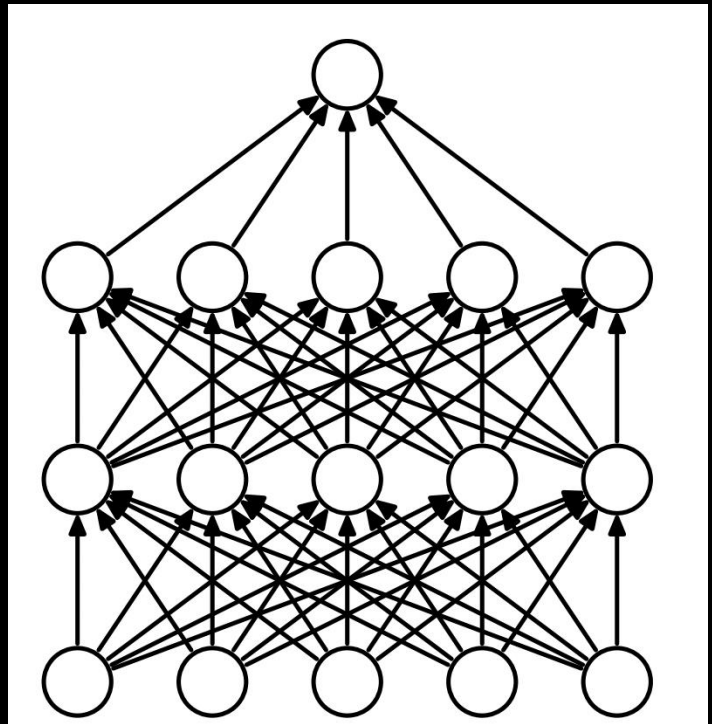
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Dropout: Approach

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Ensemble is emulated at test time by applying the network without dropout

How to handle network's expectation for a smaller activation signal than observed at test time (e.g., input from 2 versus 5 neurons)?
- Multiply each unit's outgoing weights by probability of dropping at training

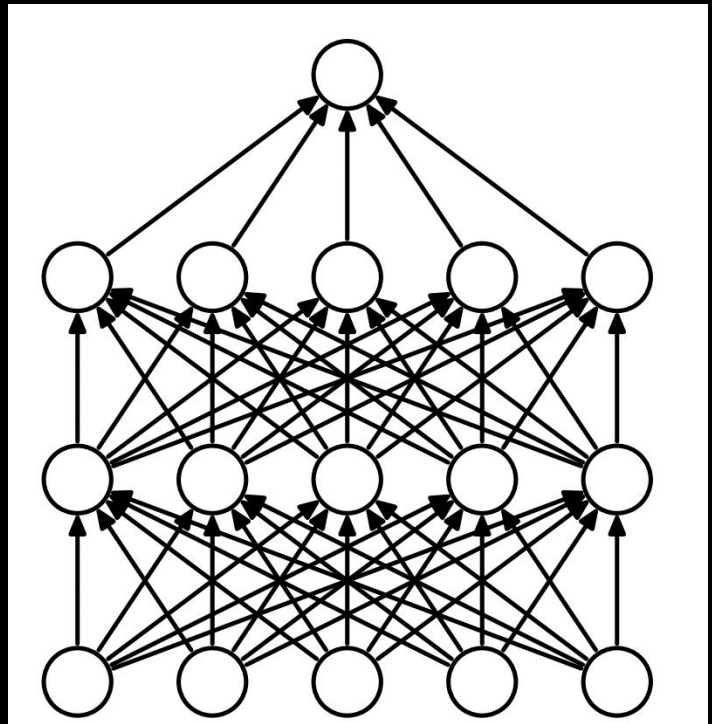


(b) After applying dropout.

Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

Dropout: Dropout vs Bagging

- Dropout approximates bagging with many models inexpensively
  - Trains algorithm repeatedly on random subsets of the training set
- Dropout differences are that subnetworks are not:
  - Trained to convergence (instead, trained for one step)
  - Independent (instead, they all share parameters)

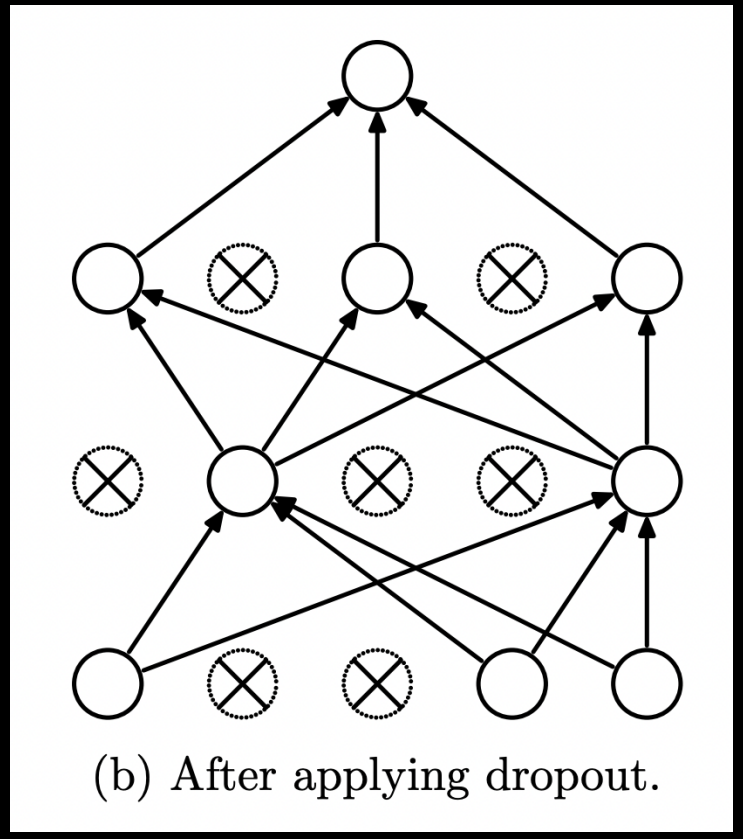


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Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

Dropout: Motivation

This approach was motivated by the role of sex in evolution: "… the role of sexual reproduction is not just to allow useful new genes to spread throughout the population, but also to facilitate this process by reducing complex co-adaptations that would reduce the chance of a new gene improving the fitness of an individual."



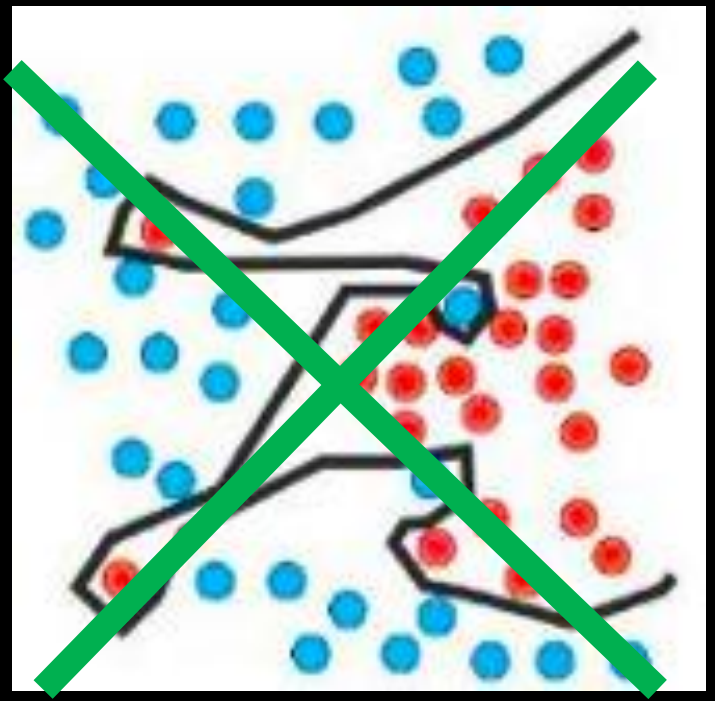
(b) After applying dropout.

"Similarly, each hidden unit in a neural network trained with dropout must learn to work with a randomly chosen sample of other units. This should make each hidden unit more robust and drive it towards creating useful features on its own without relying on other hidden units to correct its mistakes."

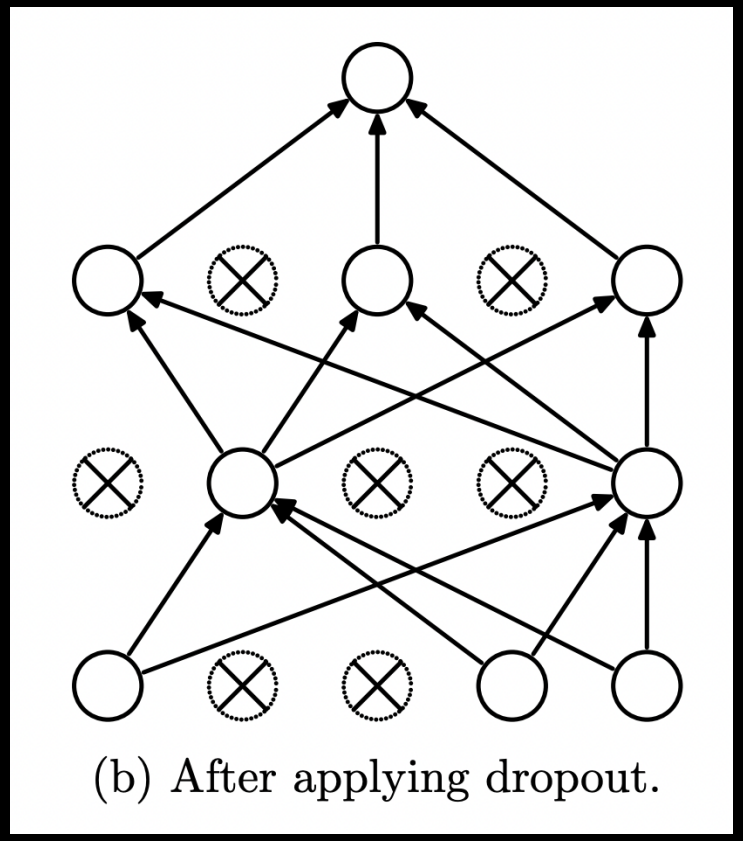
Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

Dropout: Motivation

Neurons learn to be useful alone, rather than in partnership with other neurons; e.g., mitigates large positive and negative weights canceling each other (a sign of overfitting)



https://towardsdatascience.com/techniques-for-handling-underfitting-and-overfitting-in-machine-learning-348daa2380b9



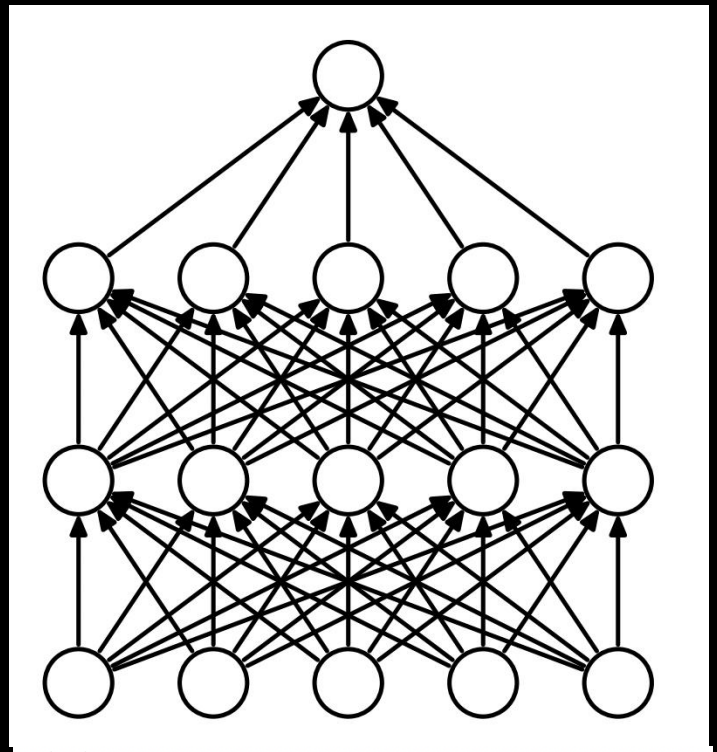
(b) After applying dropout.

Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

Dropout: Alternative Approach

A generalization of zeroing units is to instead multiply units by noise

Relevant articles:
*https://towardsdatascience.com/dropout-on-convolutional-layers-is-weird-5c6ab14f19b2
*Wu and Gu. “Towards dropout training for convolutional neural networks.” Neural Networks, 2015.

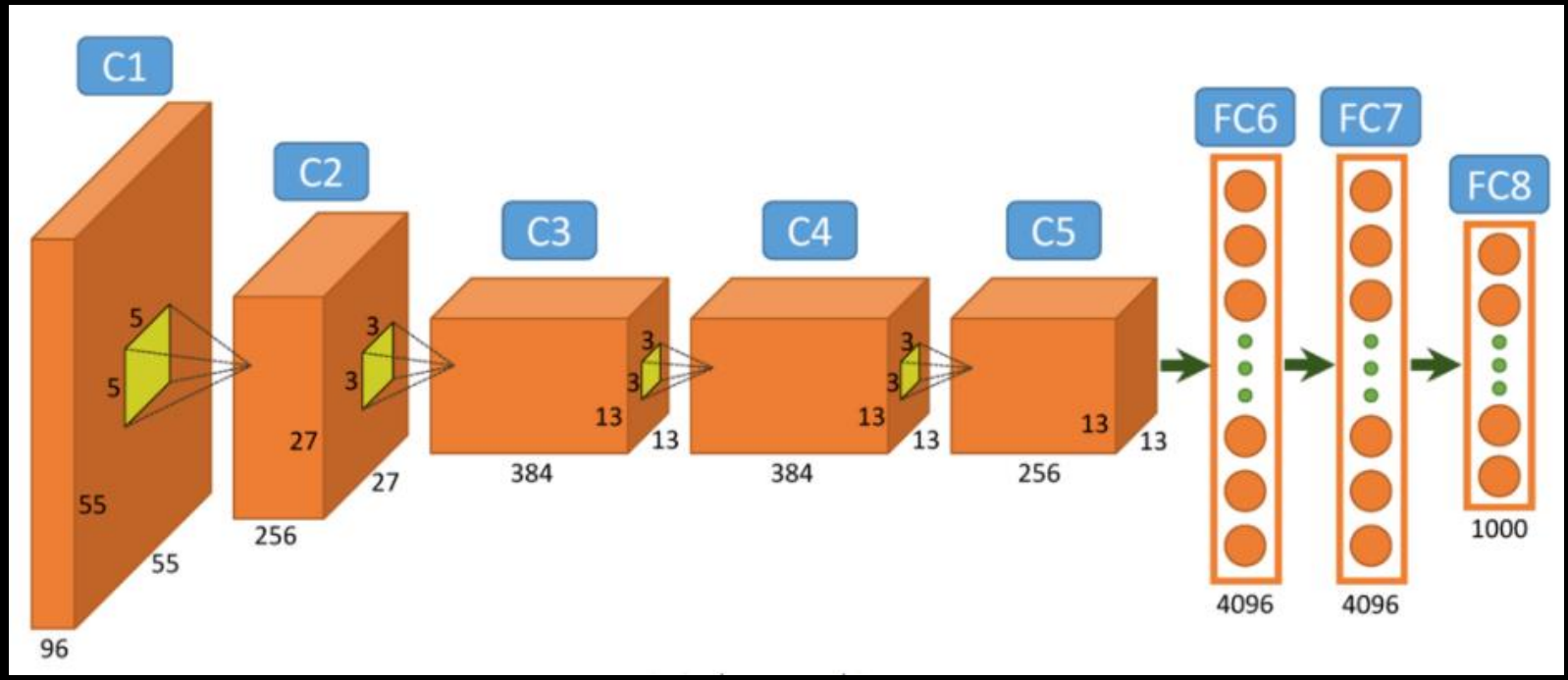


(b) After applying dropout.

Srivastava et al. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. Journal of Machine Learning Research. 2014

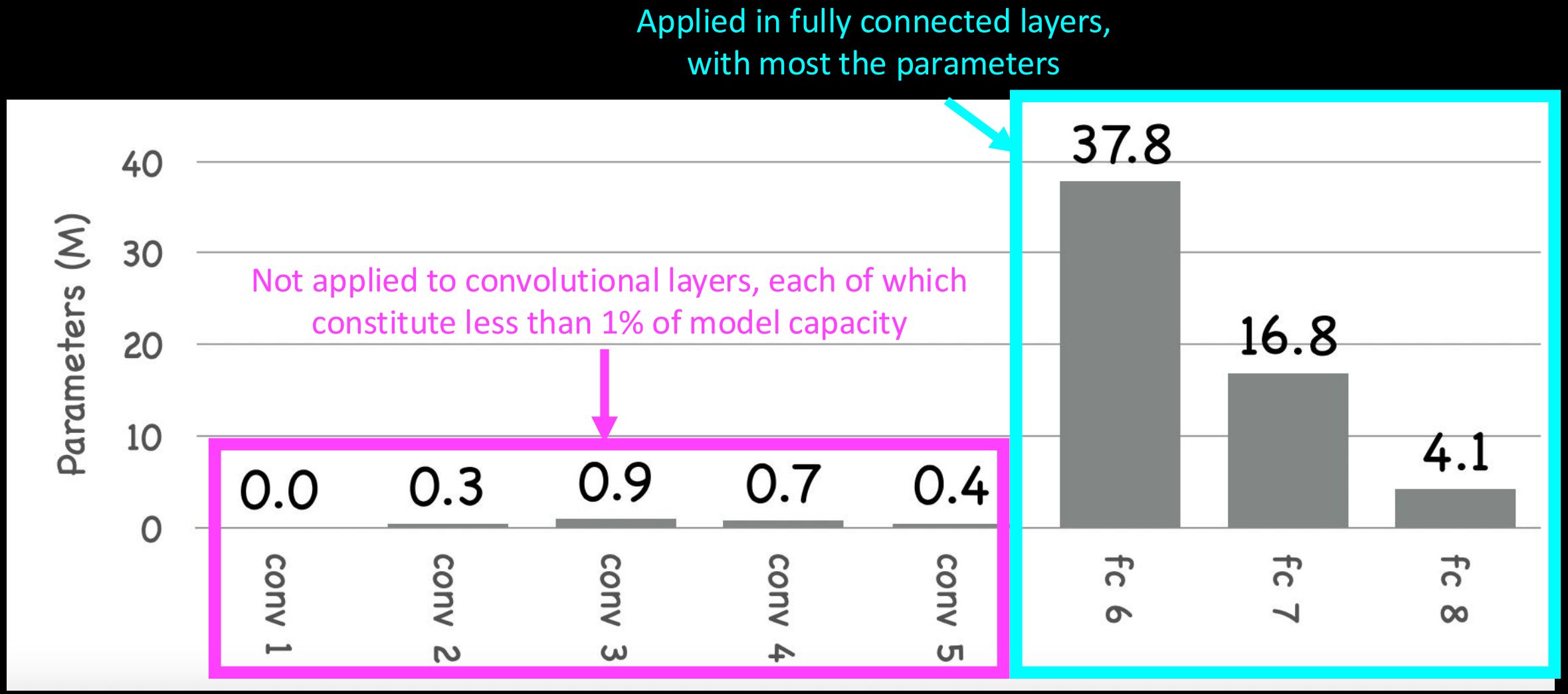
Dropout: Implementation

- Only used in fully connected layers
- Why not use it in convolutional layers?
  - Parameter tying already reduces parameter count and so offers enough regularization



https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png
https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

Dropout: Implementation

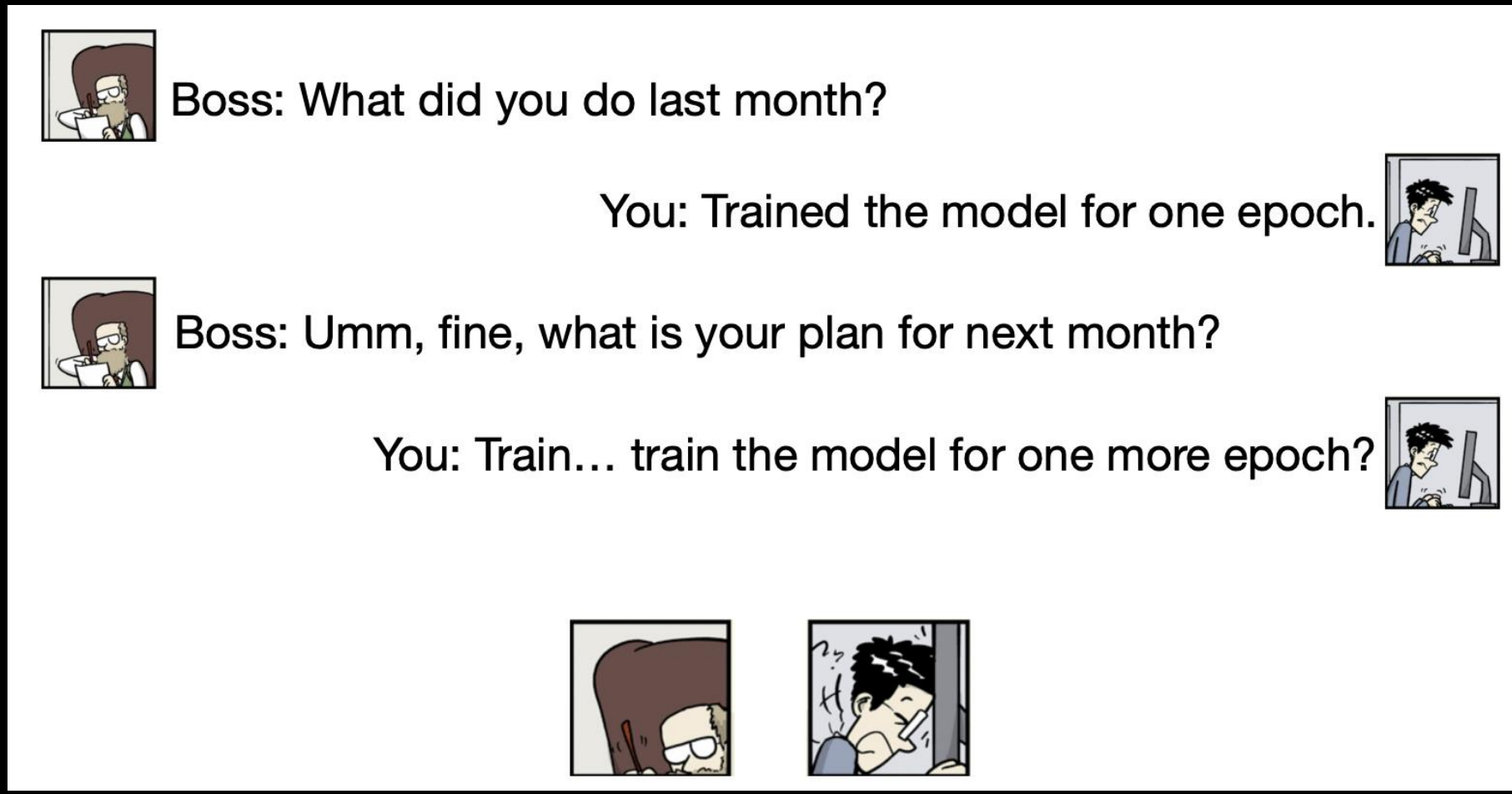


http://www.philkr.net/cs342/lectures/computer_vision/02.pdf

AlexNet Training: Key Ideas For Deep Learning

- Enable learning: use non-saturating activation functions
- Prevent overfitting: incorporate regularization methods
- Make training feasible: speed it up with better hardware

With 60M Parameters, AlexNet Was Very Time-Consuming to Train



https://hanlab.mit.edu/files/course/slides/MIT-TinyML-Lec13-Distributed-Training-I.pdf

Training: Used **Shallow Learning Techniques**

- Parameter Initialization: weight values drawn from zero-mean Gaussian distribution with standard deviation 0.01
- Batch size: 128 examples
- Weight decay: 0.0005
- Momentum optimization: manually adjusted learning rate 3 times from initial value of 0.01, dividing it by 10 each time validation error stopped falling

Speed-Up: Computations Run on 2 GPUs to Accelerate Training to 5-6 Days (for ~90 Epochs)



This image is licensed under CC-BY 2.0

Central Processing Unit (CPU)



This image is in the public domain

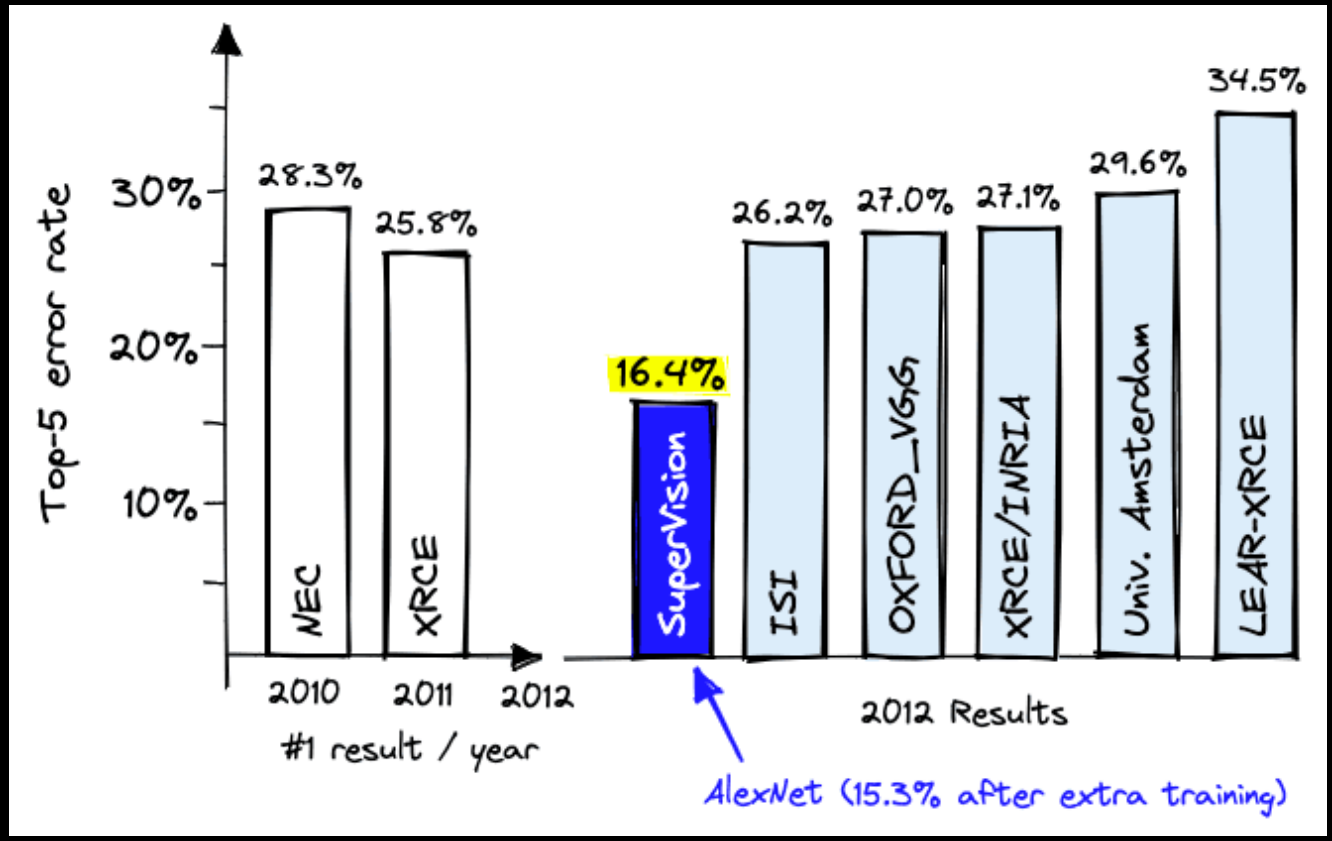
Graphics Processing Units (GPUs)



Recap: Key Ideas for Deep Learning

- Enable learning: use non-saturating activation functions (ReLUs)
- Prevent overfitting: incorporate regularization methods (data augmentation and dropout)
- Make training feasible: speed it up with better hardware (GPUs)

Experiments: Deep Learning Dominated!



Achieved unexpected, unprecedented improvements on ImageNet

(9.6 percentage point drop in top-5 error to 16.4% compared to 2011's best model)

https://www.pinecone.io/learn/series/image-search/imagenet/

Analysis: Errors

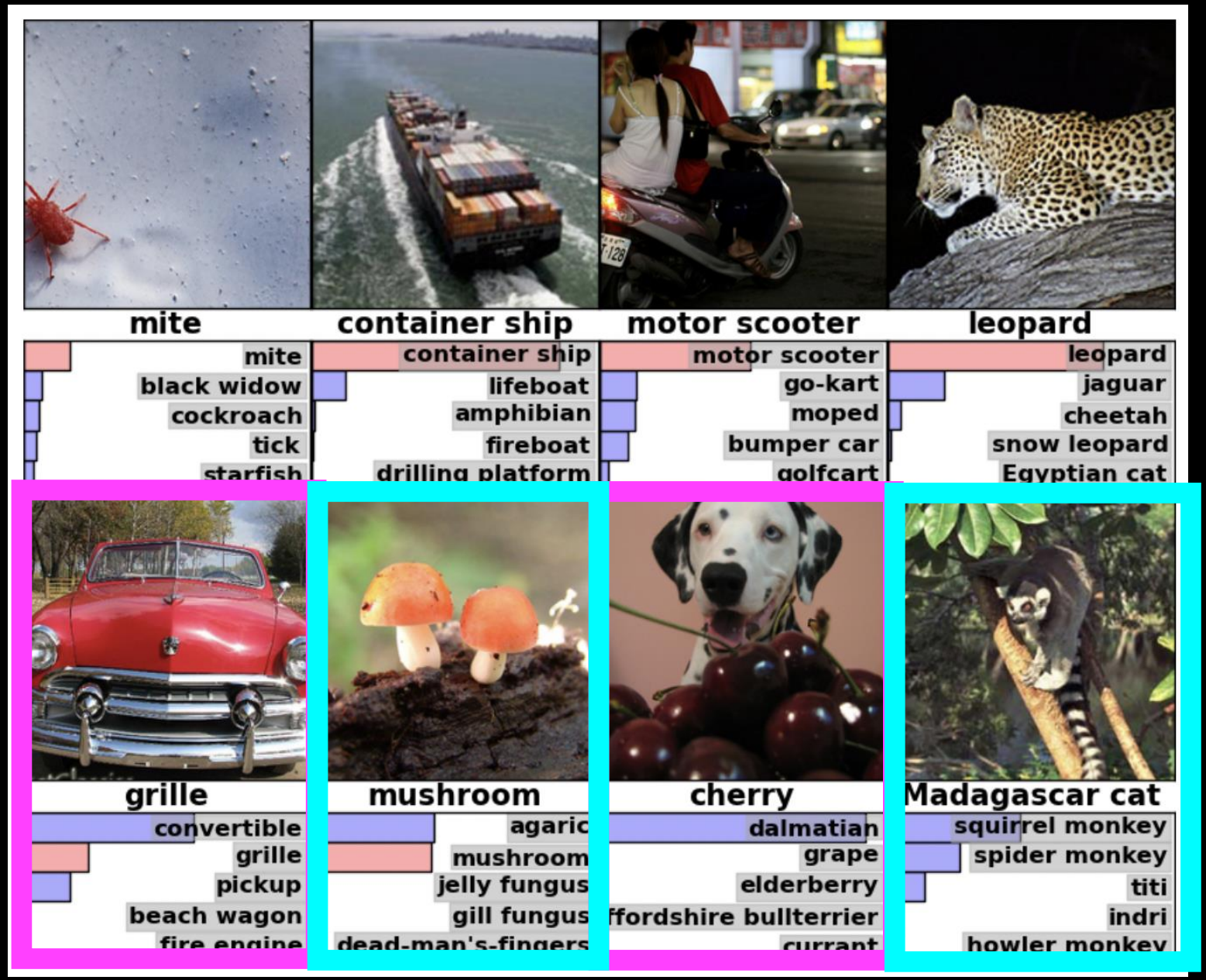
8 examples of predictions, correct and incorrect

When/why might it **succeed**?

- Single well-defined object (even if off-centered)

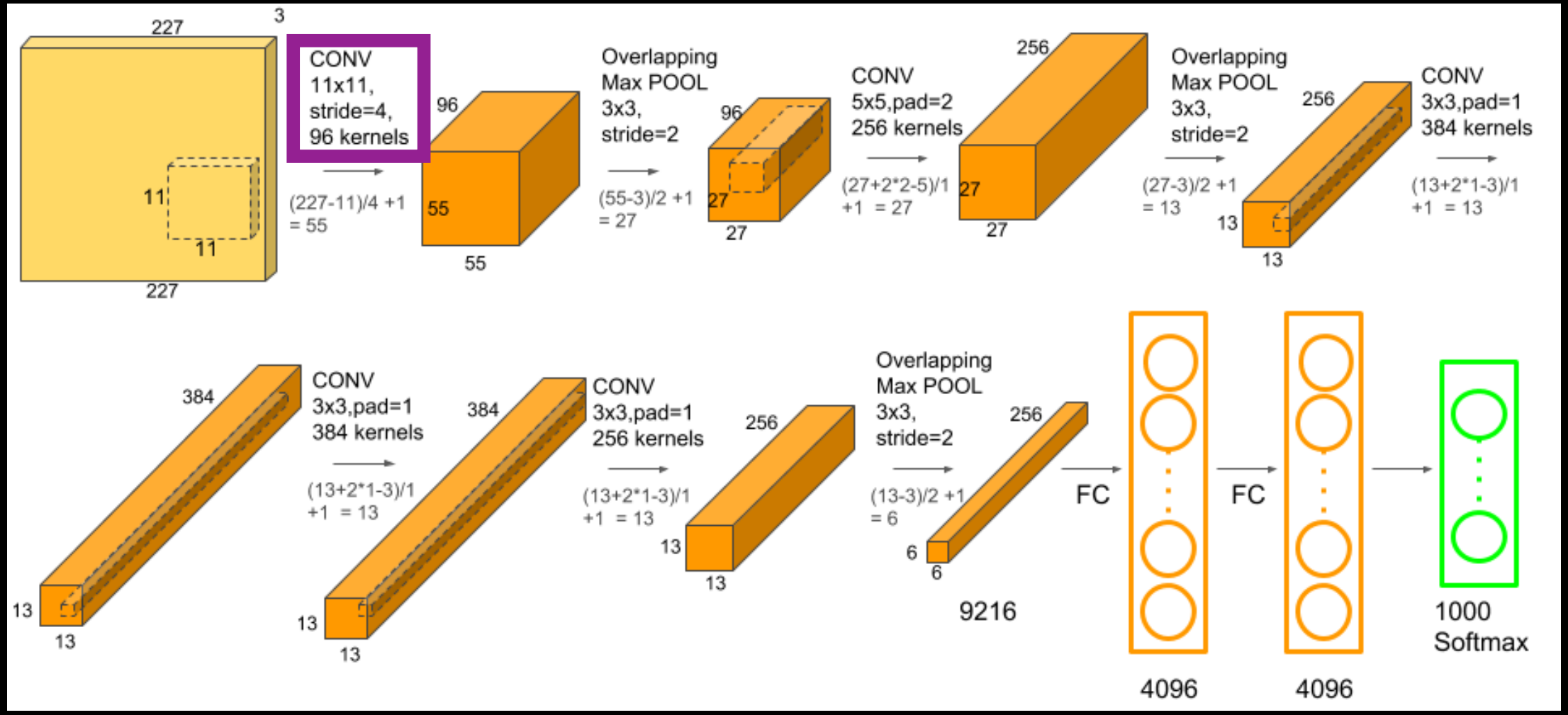
When/why might it **fail**?

- Ambiguity
- Similar categories



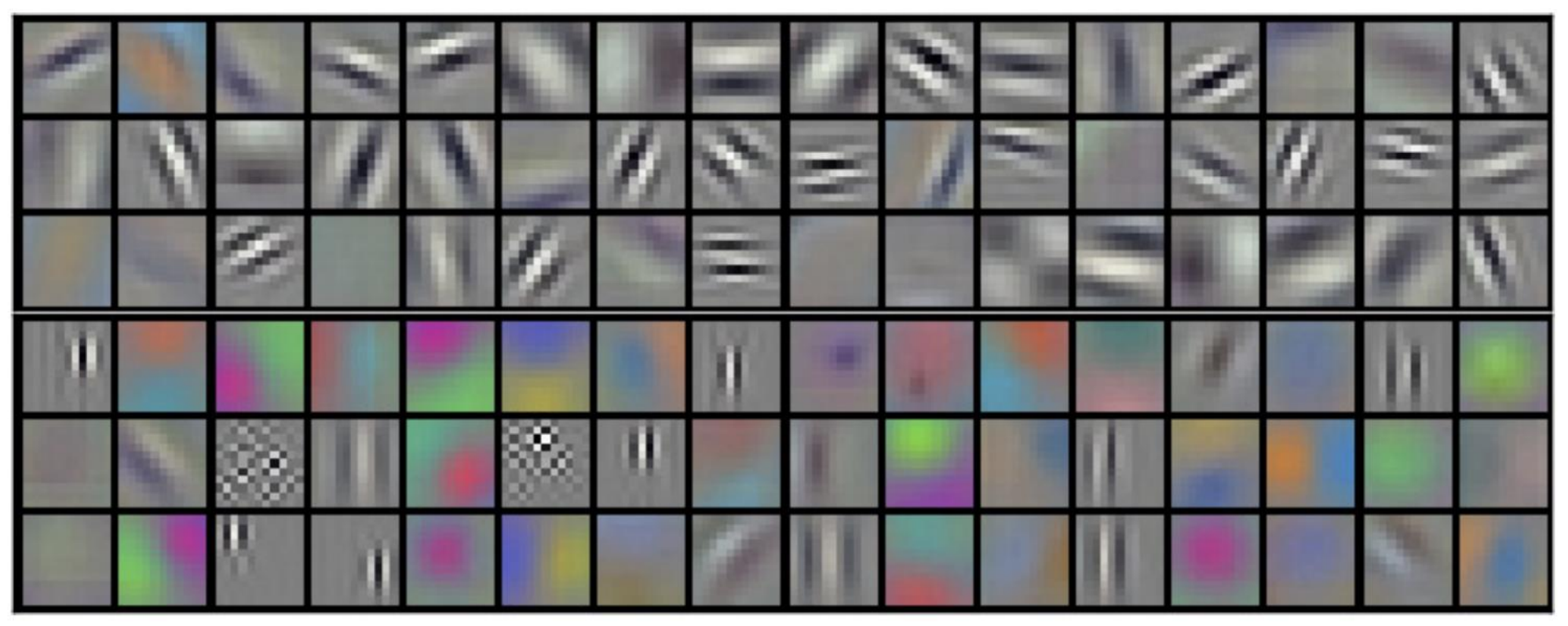
Krizhevsky et al. ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 2012.

Analysis: What Was Learned?



https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png

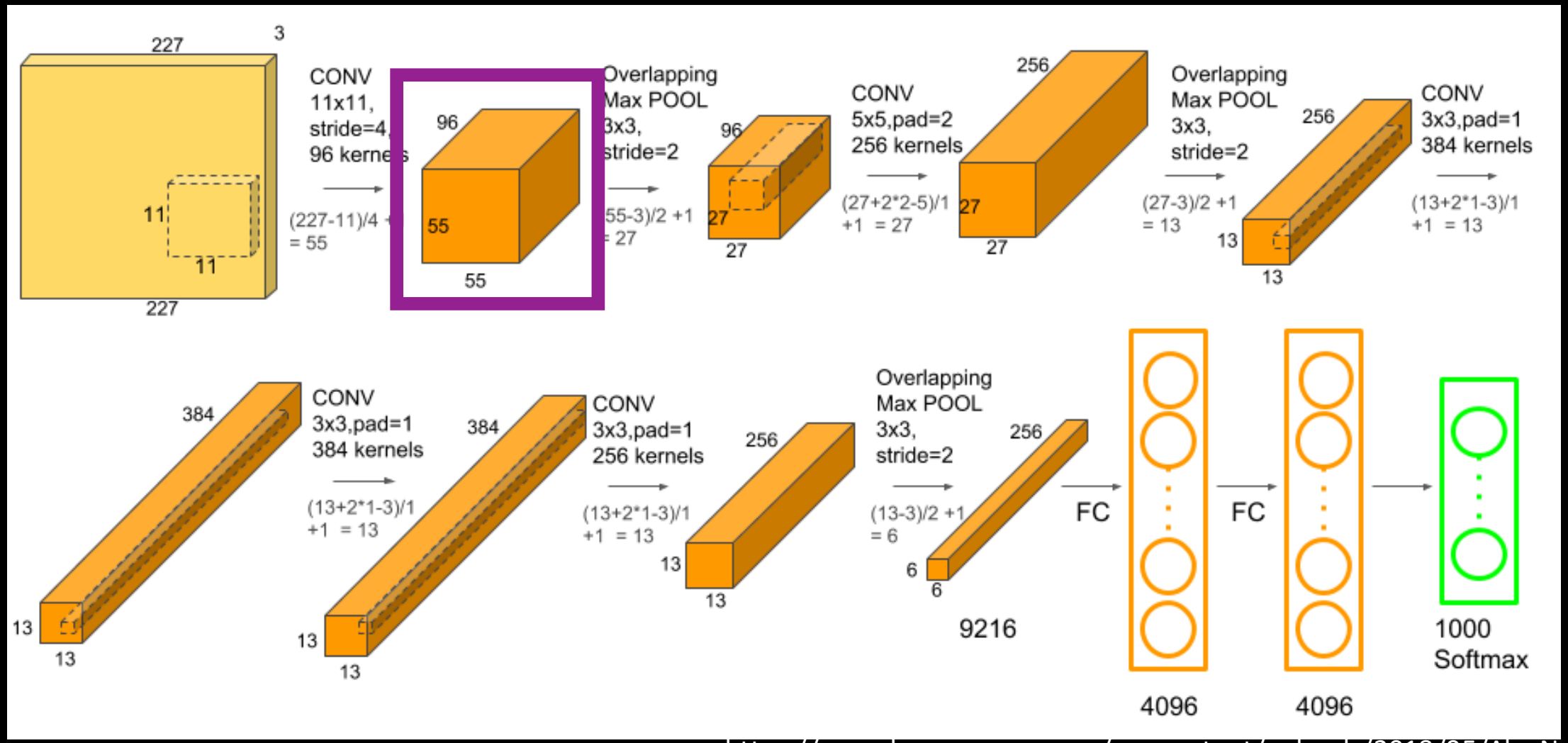
Analysis: What Was Learned? (96 Filters)



Learned model filters select based on frequency, orientation, and color!
(aligns with Hubel & Weisel's findings for how vision systems work)

Krizhevsky et al. ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 2012.

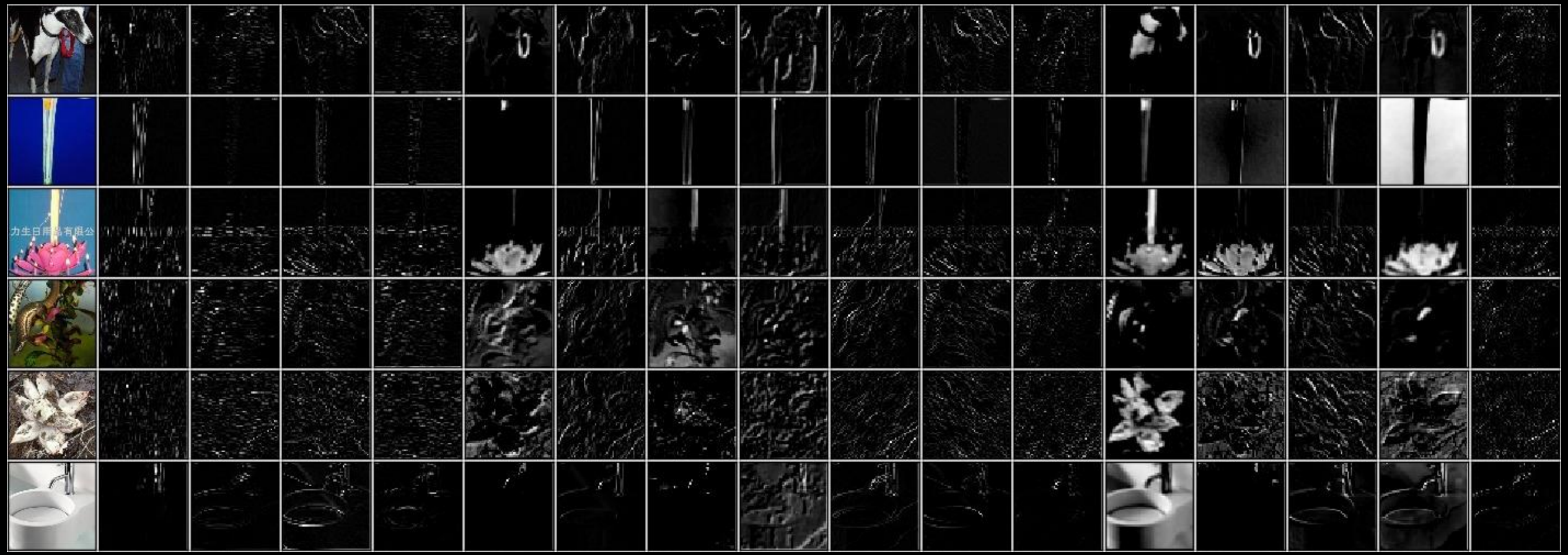
Analysis: What Was Detected?



https://www.learnopencv.com/wp-content/uploads/2018/05/AlexNet-1.png

Analysis: What Was Detected?

Images



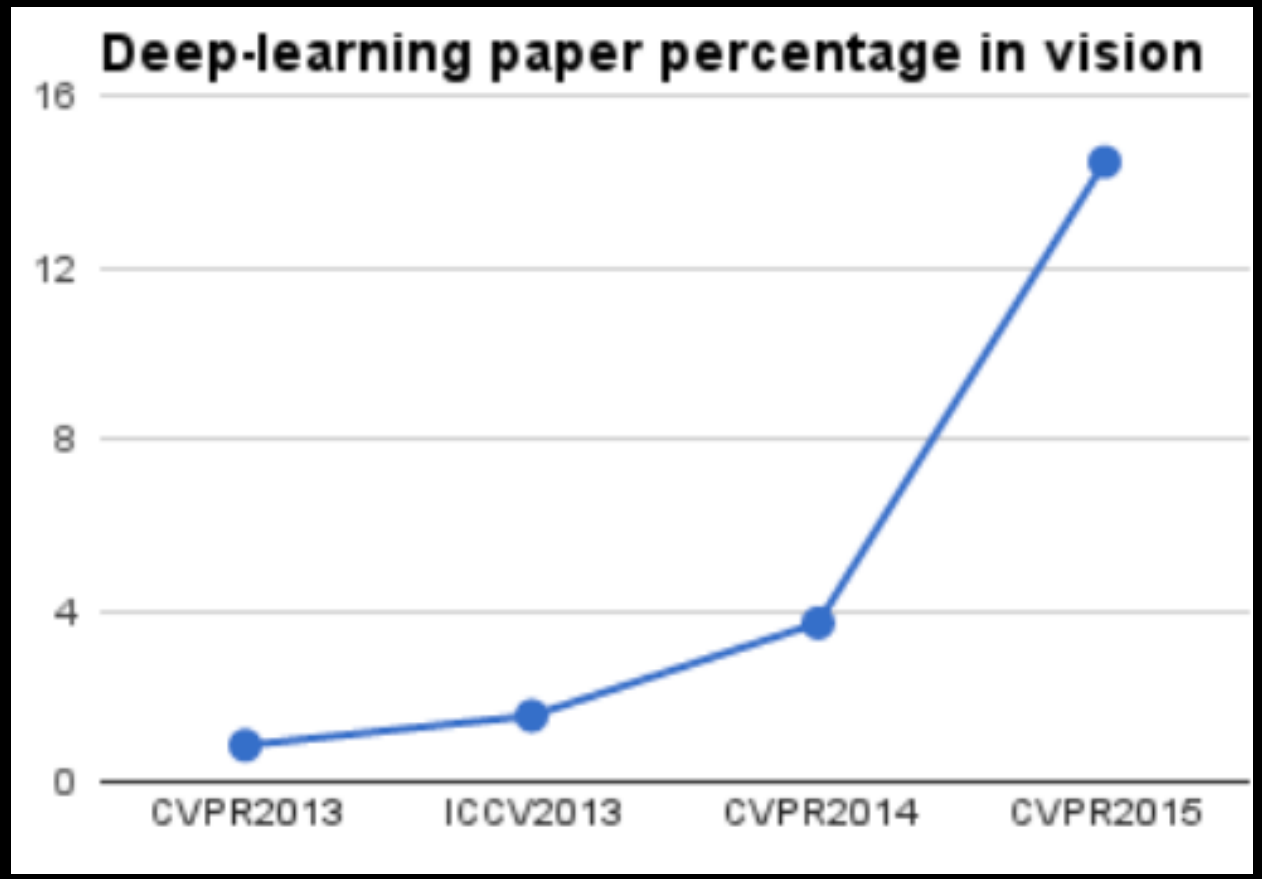
Features maps reveal that frequencies, orientations, and colors are detected

Krizhevsky et al. ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 2012.

Analysis: Influence of Depth

"… our network's performance degrades if a single convolutional layer is removed… the depth really is important for achieving our results."

Impact: Catalyzed Today's Deep Learning Revolution



Inspired, many more researchers focused on neural networks for many more problems (e.g., within the computer vision community)!

https://www.slideshare.net/xavigiro/saliency-prediction-using-deep-learning-techniques

2018 Retrospective Acknowledgment of Pioneers for Deep Learning:

Turing Award Won by 3 Pioneers in Artificial Intelligence

Share full article



From left, Yann LeCun, Geoffrey Hinton and Yoshua Bengio. The researchers worked on key developments for neural networks, which are reshaping how computer systems are built. From left, Facebook, via Associated Press; Aaron Vincent Elkaim for The New York Times; Chad Buchanan/Getty Images

https://www.nytimes.com/2019/03/27/technology/turing-award-ai.html







https://www.youtube.com/watch?v=HzilDIhWhrE&t=24s

Today's Topics

- NN neglect: 2nd AI winter & retreat to shallow models with engineered features
- NN hope lingers: the unfashionable quest for *depth*
- NNs take over: AlexNet triggers a revolution where 'deep learning' dominates
- Programming tutorial: AlexNet in PyTorch

Today's Topics

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The End